Emergent Equilibrium and the role of Expectations in a Dynamic Macroeconomic Model with Weak Complementarities

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Abstract

We propose a reduced form model of accumulation and interactions between agents. The model can be seen either as an Agent-Based model or as the DSGE one, depending on how forward-looking the agents are. When complementarities between agents are increased (but stay weak in the sense that there are no multiple period-t equilibria), non trivial dynamic properties can emerge, depending on three mechanisms: the sign of the effect of past accumulation for current actions, the strength of sluggishness in actions and the strength of the forward looking behaviour of economic agents. Depending on the relative strength of those three forces, hysteresis, limit cycles of sunspot equilibria can emerge. Our results show that complementarities in actions are key for the emergence of fluctuations in otherwise stable and non-oscillating economies.

Key Words: Agent Based Model, Rational Expectation, Coordination, Hysteresis, Limit Cycles, Sunspots;

JEL Class.: E3, E7

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Introduction

The main challenge facing macroeconomics is to understand how a large set of decentralized decisions interact as to create aggregate outcomes characterized by booms, busts, recessions, periods of exuberance and periods of despair. This is particularly challenging as we like to think of human beings as typically disliking fluctuations in their economic and social conditions. If an individual, call him Robinson, were to be secluded on an island, it is unlikely he would take decisions which would endogenously induce boom and bust cycles. The only reason there would be boom-bust cycles on a self-sufficient island is if nature imposed busts through droughts and the such.

The question of the qualitative difference between individual behaviours and aggregate outcomes has been addressed in various scientific fields. As the physicist Anderson [1972] wrote it, “More is different”, meaning that multi-component physical systems can exhibit macroscopic behaviour that cannot be understood from the laws that govern their microscopic parts, a feature known as emergent behaviour. Emergent structures can be found in many natural phenomena: hurricanes emerge from mutual positive feedback between wind, humidity, evaporation of warm surface waters and Coriolis effects. The development and growth of complex crystals or the sand dunes patterns are other examples. Swarm behaviour of marching locusts, schooling fish and flocking birds are famous natural life examples of emergent phenomena.

While there is a huge literature in many fields that studies emergent phenomena, in economics there has been little effort at understanding emergent phenomena in a systematic way. The existence of a two-way feedback between microstructure and macrostructure has been recognized within economics for a very long time, from the invisible hand of Smith [1776] to the work of Hayek [1948] and Schelling [2006]. There are many examples in economics where aggregate outcomes may display instability when individual level decisions are aimed at favouring stability. However, it is unclear what are the key forces driving such phenomena.

How did modern macroeconomics addressed the issue of macroeconomic fluctuations? Dynamic Stochastic General Equilibrium models[1] typically see the macroeconomy as being

[1] See Kydland and Prescott [1982] for pioneering work and Smets and Wouters [2007] for more recent New-Keynesian incarnation.

smoothly converging to a stable steady state absent of shocks, and fluctuations are created by exogenous shocks to fundamentals. In that sense, there is no qualitative jump between individuals and the aggregate in the most standards incarnations of those models. Although the steady state is stable absent of shocks, there is a possibility of belief shocks (sunspots) to affect the economy when agent have rational expectations and the equilibrium is indeterminate, as introduced by Cass and Shell [1983], Azariadis [1981] and Benhabib and Farmer [1994]. The assumption of rational expectations is then crucial to generate such emergent self-fulfilling fluctuations.

Agent Based Models[2] is another route taken by macroeconomists to bridge the microeconomic behaviours and the macroeconomic outcomes. In these models, autonomous agents are interacting with bounded rationality and not fully rational expectations, and the aggregate properties of the model cannot be directly inferred from micro behaviour, leading to the emergent phenomena such as boom-bust cycles and hysteresis.

In this paper, we study what we believe is the minimal model that one can write to generate emergent aggregate outcomes. It features accumulation, interactions between agents and rational expectations. Absent of the expectation part, it can be thought as a minimal agent-based model. With the expectation part, it can be thought as a minimal reduced form of a DSGE model[3].

In both cases, we study how the strength of agents interactions affect aggregate outcomes, and make explicit the conditions under which the aggregate economy looks similar to individual behaviours and when it displays emergent phenomena such as hysteresis, endogenous cycles and self-fulfilling equilibria.

The paper is organized as follows. In a first section, we will first consider the backward version of the model (the "ABM version"). In a second section, we will consider the model with rational expectations (the "DSGE version"). Our main results are that hysteresis and limits cycles exist on rbtoh versions of the model at business cycle frequency, while self-fulfilling equilibria (obviously) rely crucially on the forward-looking part of the model.

[2]See Schelling [1971] for a very first application to social sciences and Delli Gatti, Fagiolo, Gallegati, Richiardi, and Russo [2018] for an account of Agent Based modelling in economics.

[3]See Beaudry, Galizia, and Portier [2015] and Beaudry, Galizia, and Portier [2020] for a formal derivation in two different settings

Our conclusion is that cycles and hysteresis are robust emergent phenomena, that occur in both the backward and forward-looking configurations of the model when one increases complementarities.

1 The Backward-Looking Model

1.1 Baseline Model: No Interactions

The environment we consider is one with a large number N of agents indexed by i, where each agent makes effort e_{it}, which accumulates into X_{it} (X_{it} can represent either physical capital or a durable consumption good). The accumulation equation is given by

$$X_{it+1} = (1-\delta)X_{it} + e_{it}\,. \qquad 0 < \delta < 1\,. \tag{1}$$

Suppose initially that there is no interaction between agents and that the decision rule for agent i's effort is given by

$$e_{it} = \max\{\alpha_0 - \alpha_1 X_{it} + \alpha_2 e_{it-1}, 0\}\,, \tag{2}$$

where $\alpha_0 > 0$, $-1 < \alpha_1 < 1$, and $0 < \alpha_2 < 1$. In this decision rule, the effect of X_{it} on effort can be negative ($\alpha_1 > 0$) so as to reflect some underlying decreasing returns to capital accumulation, or positive ($\alpha_1 < 0$) so as to reflect that a higher asset level can increase the productivity of effort. Finally, the effect of the previous period's action is positive so as to reflect sluggish response due for example to adjustment costs. Since the steady state(s) of this model will be non-zero, and since we will largely be concerned with local dynamics around a particular steady state (SS), in practice we will often ignore the non-negativity constraint captured by the max operator in this decision rule wherever doing so is justified.

We will always restrict the analysis to symmetric allocations, so that $e_{it} = e_t$ and $X_{it} = X_t$. Ignoring the non-negativity constraint, the aggregate dynamics of the economy are then given by the linear system:

$$\begin{pmatrix} e_t \\ X_t \end{pmatrix} = \underbrace{\begin{pmatrix} \alpha_2 - \alpha_1 & -\alpha_1(1-\delta) \\ 1 & 1-\delta \end{pmatrix}}_{M_L} \begin{pmatrix} e_{t-1} \\ X_{t-1} \end{pmatrix} + \begin{pmatrix} \alpha_0 \\ 0 \end{pmatrix}\,. \tag{3}$$

We make the following assumption.

Assumption A1. *The parameters satisfy:*

(i) $\alpha_1 > -(1-\alpha_2)\delta$,

(ii) $\alpha_1 < \left(\sqrt{\alpha_2} - \sqrt{1-\delta}\right)^2$.

Letting $\rho^* \equiv 1 - \alpha_2 + \alpha_1/\delta$, Assumption A1(i) implies $\rho^* > 0$, and therefore the unique SS of the model is given by $SS_0 = (e^s, X^s)$, where $e^s = \alpha_0/\rho^* > 0$ and $X^s = e^s/\delta$. The stability properties of (3) are established in Proposition 1.

Proposition 1. *Under Assumption A1, both eigenvalues of the matrix M_L are real, positive and inside the unit circle. Therefore, the unique steady state is stable, and the system converges to it without cycles.*

All proofs are given in the appendix. According to Proposition 1, the dynamics are extremely simple, with the system converging to SS_0 for any starting values of $X_{it} = X_t$ and $e_{it-1} = e_{t-1}$ (see Figure 1). We now add interactions to the model and study how is the dynamics affected.

1.2 Adding Interactions

We extend the previous setup to allow for interactions between individuals by having the investment rule be given instead by

$$e_{it} = \max\left\{\alpha_0 - \alpha_1 X_{it} + \alpha_2 e_{it-1} + F\left(e_t\right), 0\right\} \tag{4}$$

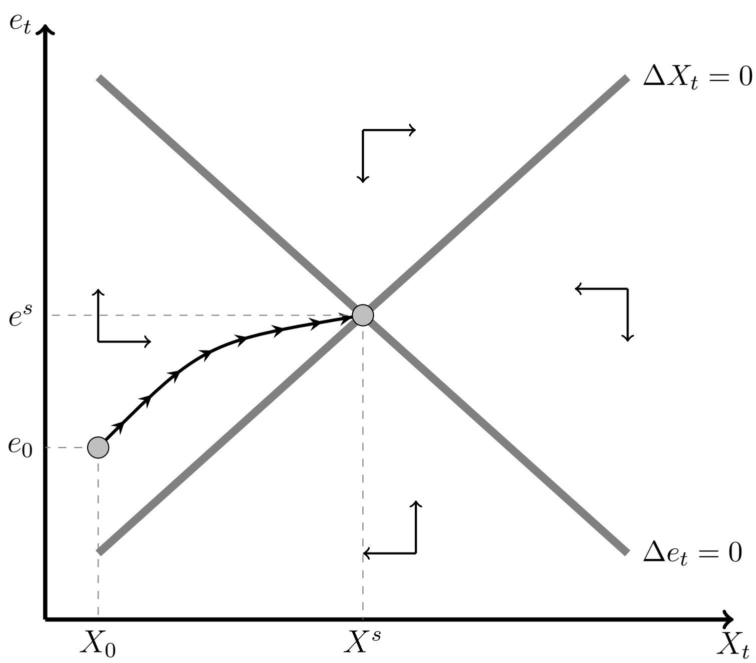
with $e_t = \frac{\sum e_{jt}}{N}$, while the law of motion of X (equation (1)) is kept unchanged. As before, since the SSs will be strictly positive and we are largely interested in the local dynamics around a SS, we will ignore the non-negativity constraint captured by the max operator wherever doing so is justified. In a symmetric equilibrium, the system can then be written

$$\begin{aligned} e_t - F\left(e_t\right) &= \alpha_0 - \alpha_1 X_t + \alpha_2 e_{t-1}\,, \\ X_{t+1} &= \left(1-\delta\right) X_t + e_t\,. \end{aligned} \tag{5}$$

1.2.1 The interaction function

The function F embeds social interactions between agents. In a fully specified model, those interactions could be mediated through prices, market thickness, or externalities, among

Figure 1: Phase Diagram in the Model without Interactions



This figure represent the phase diagram of the dynamic system (3), in which there are no interactions, and when A1 holds.

other things. We assume that agents take the average action in the economy as given, so that (4) can be interpreted as the best-response rule of an individual to the average action. We make a number of assumptions about the function F. The set of assumptions combined here into Assumption A2 is not overly restrictive and allows for a cleaner characterization of the model dynamics.

Assumption A2. *The interaction function F is given by $F(e) = \rho\widetilde{F}(e)$, where $\widetilde{F}$ belongs to the following family of functions:*

(i) $\widetilde{F}$ *is three times continuously differentiable;*

(ii) $\widetilde{F}(e^s) = 0$, *where e^s is the steady state of the model without interactions;*

(iii) $\widetilde{F}(0) > -\alpha_0$;

(iv) $0 < \widetilde{F}'(e) \leq 1$, *with* $\widetilde{F}'(e^s) = 1$;

(v) $\lim_{e\to 0} \widetilde{F}'(e) = \lim_{e\to\infty} \widetilde{F}'(e) = 0$;

(vi) $\widetilde{F}''(e) > 0$ *on* $e < e^s$ *and* $\widetilde{F}''(e) < 0$ *on* $e > e^s$, *with* $\widetilde{F}'''(e^s) < 0$.

The function $\widetilde{F}$ here is a strictly increasing (possibly unbounded) sigmoid function whose slope attains a unique maximum of 1 at $e = e^s$. For $\rho \neq 0$, the interaction function F inherits these properties, except (a) F is decreasing instead of increasing if $\rho < 0$ (i.e., F is an inverted sigmoid), and (b) the maximum value of $|F'(e)|$ is $|\rho|$ instead of 1. Thus, the type of strategic interaction is governed by the sign of ρ (complementarity if $\rho > 0$, substitutability of $\rho < 0$), while the strength of those interactions is assumed to be strongest at $e = e^s$, becoming monotonically weaker as e moves away from e^s in either direction.

The parameter ρ simultaneously controls both the type of strategic interaction (substitutability vs. complementarity), as well as the maximum strength of those interactions. The following assumption imposes the additional condition that if there are complementarities ($\rho > 0$ case) then those complementarities are always weak, in the sense that $F'(e) < 1$ everywhere.

Assumption A3. $\rho < 1$.

1.2.2 Period-t Equilibrium

In a given period t, the (symmetric) equilibrium is given by $e_t = \alpha_0 - \alpha_1 X_t + \alpha_2 e_{t-1} + F(e_t)$, which is indexed by the history as summarized by $\alpha_t = \alpha_0 - \alpha_1 X_t + \alpha_2 e_{t-1}$.

Proposition 2. *Under Assumptions A1-A3, there exists a unique period-t equilibrium.*

In Figure 2, for the case where $\alpha_t + F(0) > 0$, we illustrate graphically the determination of the period-t equilibrium with, respectively, substitutabilities, strong complementarities ($F'(e) > 1$ for some values of e), and weak complementarities. Note that the case of strong complementarities violates Assumption A3.

Figure 3 shows agent i's policy function for two values of the intercept—i.e., for two different histories $\alpha_t = \alpha_0 - \alpha_1 X_t + \alpha_2 e_{t-1}$—so as to make clear its dependence on past effort decisions and on the stock variable X_t. The dynamics of the system are induced by the fact that past effort decisions determine the location of the policy function, which in turn determines the period-t equilibrium level of effort. Thus, today's decision feeds into the determination of next period's intercept.[4]

1.2.3 Steady State(s)

Under Assumption A2(ii), the SS of the model without interactions, SS_0, is also a SS of the model with interactions (regardless of the value of ρ). Recalling the definition $\rho^* \equiv 1 - \alpha_2 + \alpha_1/\delta$ from the no-interactions model, which is strictly positive by Assumption A1(i), we have the following proposition.

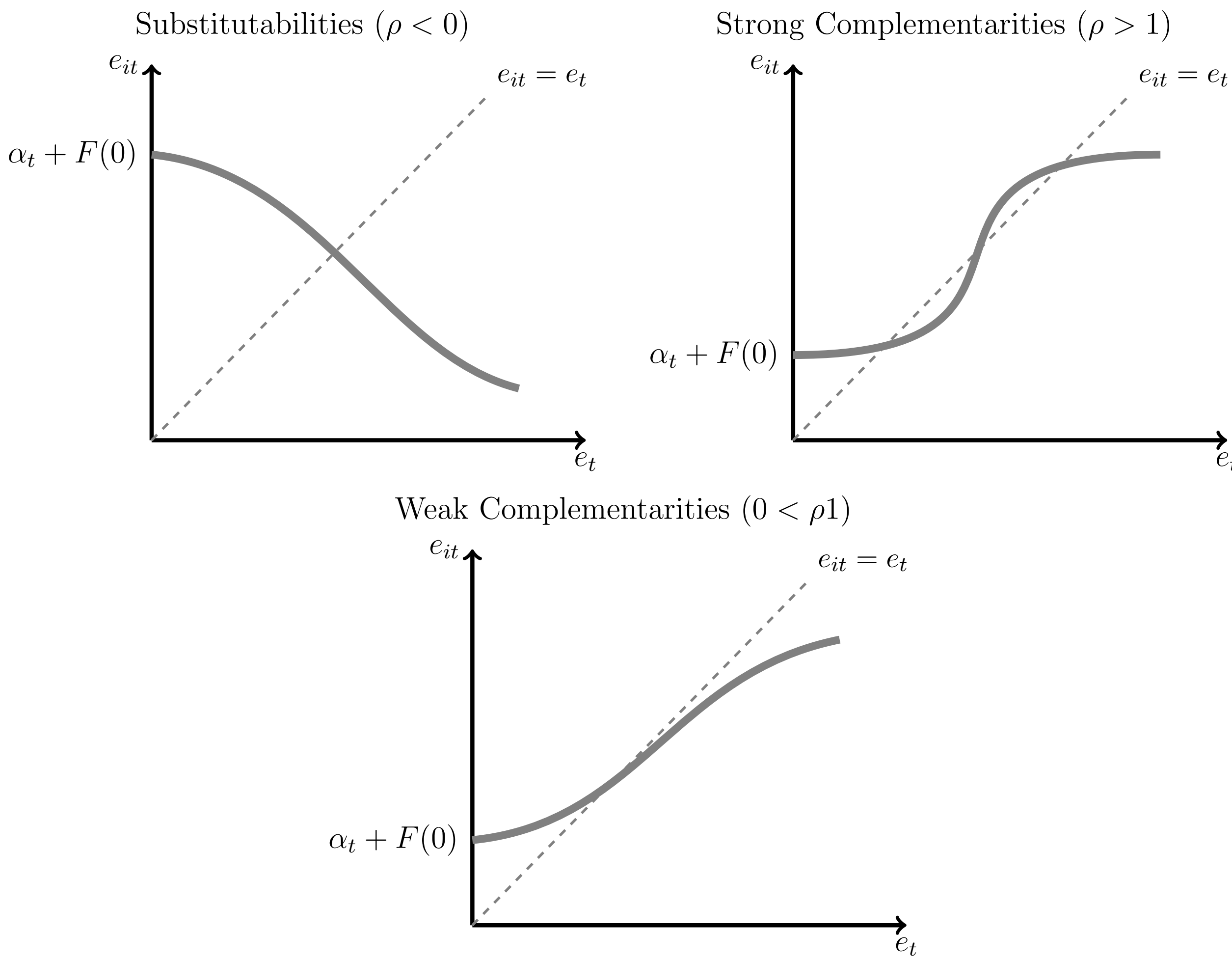
Proposition 3. *Suppose Assumptions A1-A3 hold. Then:*

(i) If $\rho \leq \rho^*$—which is necessarily the case if either $\rho \leq 0$ (substitutabilities) or $\alpha_1 \geq \delta\alpha_2$—then SS_0 is the unique steady state.

(ii) If $\rho > \rho^*$—which is only possible if both $\rho > 0$ (complementarities) and $\alpha_1 < \delta\alpha_2$—then there are two steady states in addition to SS_0 (one with $e > e^s$ and one with $e < e^s$).

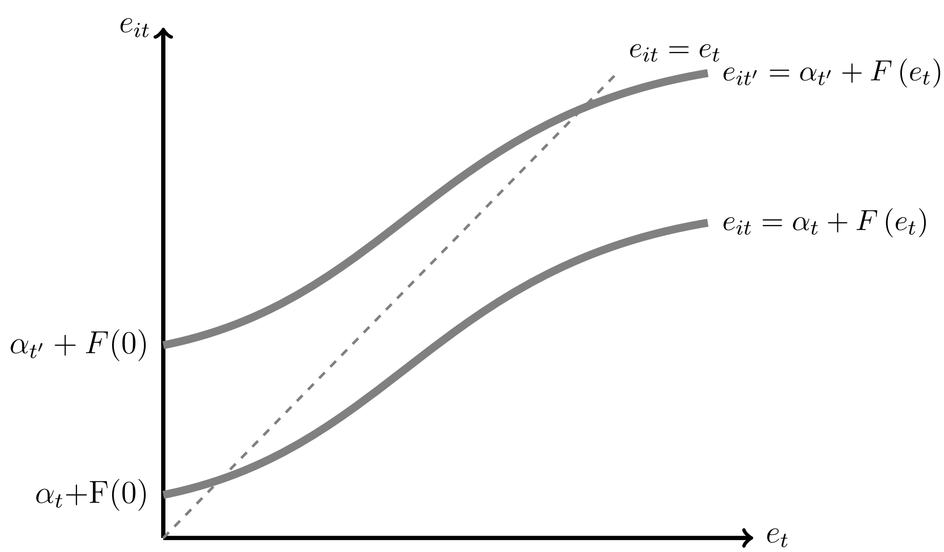
[4]Under Assumption A3, the period-t equilibrium depicted in Figure 3 is stable under a *tâtonnement*-type adjustment process. This stability property is not the focus of the current paper. Instead we are interested in the explicit dynamics induced by the system (5).

Figure 2: Period-t Equilibrium



Notes: This figure plots agent i's policy rule (equation (4)): $e_{it} = \alpha_t + F(e_t)$, with $\alpha_t = \alpha_0 - \alpha_1 X_t + \alpha_2 e_{t-1}$. Period-t equilibria are at intersections of the policy rule and the 45-degree line $e_{it} = e_t$.

Figure 3: Policy Rule for Two Different Histories with Weak Complementarities



This figure plots the best response rule (equation (4)): $e_{it} = \alpha_t + F\left(e_t\right)$, with $\alpha_t = \alpha_0 - \alpha_1 X_{it} + \alpha_2 e_{it-1}$. The intercepts α_t and $\alpha_{t'}$ correspond to two different histories of the model.

Proposition 3 establishes that, for the case of substitutabilities ($\rho < 0$), the unique SS of the system is the no-interaction one SS_0, and that, in order for multiple SSs to emerge (if they ever do), there must be sufficiently strong complementarities.

Our goal is now to use this framework to examine how the dynamics of this system are affected by the properties of the interaction effect ρ, and especially what conditions on $F(\cdot)$ will give rise to emergent equilibria.

1.3 Local Dynamics of the Model With Interactions

We now consider the dynamics given by (5). The first-order approximation of the system around SS_0 is given by

$$\begin{pmatrix} e_t \\ X_t \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{\alpha_2-\alpha_1}{1-\rho} & -\frac{\alpha_1(1-\delta)}{1-\rho} \\ 1 & 1-\delta \end{pmatrix}}_{M} \begin{pmatrix} e_{t-1} \\ X_{t-1} \end{pmatrix} + \begin{pmatrix} \left(1-\frac{\alpha_2-\alpha_1}{1-\rho}\right) e^s + \frac{\alpha_1(1-\delta)}{1-\rho} X^s \\ 0 \end{pmatrix} . \tag{6}$$

In order to understand the dynamics of the system, it is informative to first look at local dynamics in the neighborhood of this SS. The eigenvalues of the matrix M are the roots of the quadratic equation

$$Q(\lambda) = \lambda^2 - T\lambda + D = 0\,, \tag{7}$$

where T is the trace of M (and also the sum of its eigenvalues) and D is the determinant of M (and also the product of its eigenvalues). The two eigenvalues are

$$\underline{\lambda}, \overline{\lambda} = \frac{T}{2} \pm \sqrt{\left(\frac{T}{2}\right)^2 - D} \tag{8}$$

where

$$T = \frac{\alpha_2-\alpha_1}{1-\rho} + (1-\delta) \tag{9}$$

and

$$D = \frac{\alpha_2(1-\delta)}{1-\rho}\,. \tag{10}$$

Recall that $\rho = F'(e^s)$ parameterizes the type and degree of the strategic interactions at SS_0. As noted above, when there are no strategic interactions ($\rho = 0$), the roots of the system are real, positive and smaller than one. We now consider what happens to these roots as ρ is allowed to vary between $-\infty$ and 1.

Geometric analysis

It is useful to give a geometric analysis of the location of the two eigenvalues of the system. This is done in Figure 4, which represents the plane (T, D). A point in this plane is a (T, D) pair corresponding to a particular configuration of the model parameters (including ρ). We have drawn three lines and a parabola in that plane. The line passing through B and C corresponds to $D = 1$, the line through A and B to the equation $Q(-1) = 0$ ($\Leftrightarrow D = -T-1$), and the line through A and C to the equation $Q(1) = 0$ ($\Leftrightarrow D = T - 1$). On the perimeter of the triangle $\widehat{ABC}$, at least one eigenvalue has a modulus of 1. Both eigenvalues of the system are inside the unit circle when (T, D) is inside the triangle $\widehat{ABC}$, while at least one is outside the unit circle when (T, D) is outside $\widehat{ABC}$. Whether the eigenvalues are real or complex depends on the sign of the discriminant of the equation $Q(\lambda) = 0$, which is given by $\Delta = \left(\frac{T}{2}\right)^2 - D$. The parabola in Figure 4 corresponds to the equation $\Delta = 0$ (i.e., $D = T^2/4$). The eigenvalues are complex conjugates above the parabola, and real below it. Finally, the eigenvalues are both real and positive in the northeast quadrant of Figure 4, both real and negative in the northwest quadrant, and of opposite signs in the lower half of the plane.

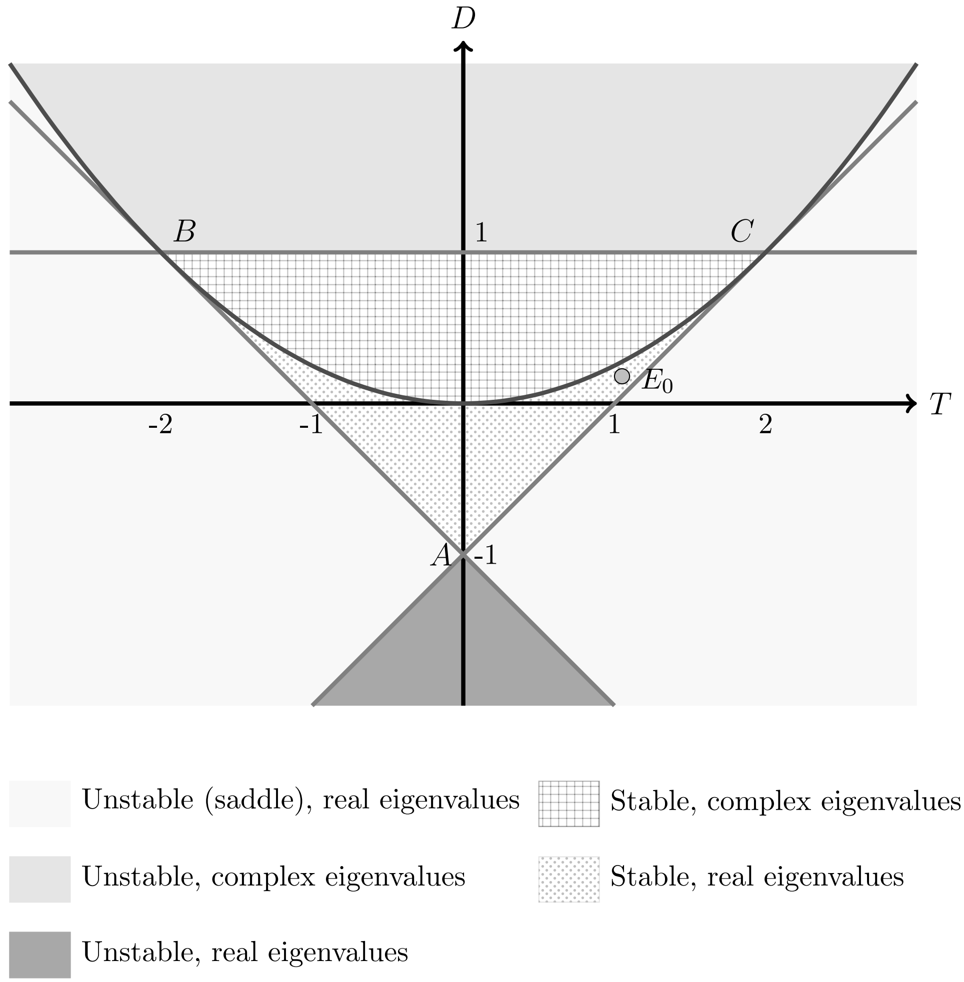
Depending on the particular configuration of the parameters, the pair (T, D) could in principle be in any of the regions shown in Figure 4, each of which corresponds to qualitatively different local dynamics around SS_0. In particular, SS_0 can be locally stable or unstable,[5] with either a pair of complex conjugate eigenvalues, or a pair of real eigenvalues each of which could be either positive or negative. When both eigenvalues are (weakly) positive we will say that the system exhibits monotonic dynamics since, given any initial position at date 0, for t large enough e_t and X_t will be monotone functions of t. On the other hand, if the system has any negative or complex eigenvalues we will say that it exhibits cycles or oscillations.

Under Assumption A1, Proposition 1 showed that, when $\rho = 0$, SS_0 corresponds to some point E_0 in the region of Figure 4 that is below the parabola (the eigenvalues are real), in the upper-right quadrant (the eigenvalues are both positive, so that their product D and

[5]Note that this system is purely backward-looking, and therefore the presence of any unstable eigenvalues implies that the system will generally diverge from the SS. With some abuse of terminology, for the sake of brevity we use "unstable" to refer to any situation with at least one unstable eigenvalue, even if there is also a stable eigenvalue (i.e., the SS is a saddle).

sum T are both positive), and to the left of the line through A and C (both eigenvalues are less than one). An example of such a point is indicated in the Figure.

Figure 4: Possible configurations of the local dynamics



This figure shows the plane (T, D), where T is the trace and D the determinant of the matrix M. The point E_0 correspond to the model without demand externalites. It is placed in the zone where the two eigenvalues are real, positive and smaller than 1.

As ρ varies, the eigenvalues of the system will generally vary, implying changes to the dynamic behavior of e and X. From equations (9) and (10), we may obtain the following relationship between the trace and determinant of M:

$$\begin{cases} D = \dfrac{\alpha_2(1-\delta)}{\alpha_2 - \alpha_1}\left[T - (1-\delta)\right] & , \text{if } \alpha_1 \neq \alpha_2 \\ T = 1 - \delta & , \text{if } \alpha_1 = \alpha_2 \end{cases} . \tag{11}$$

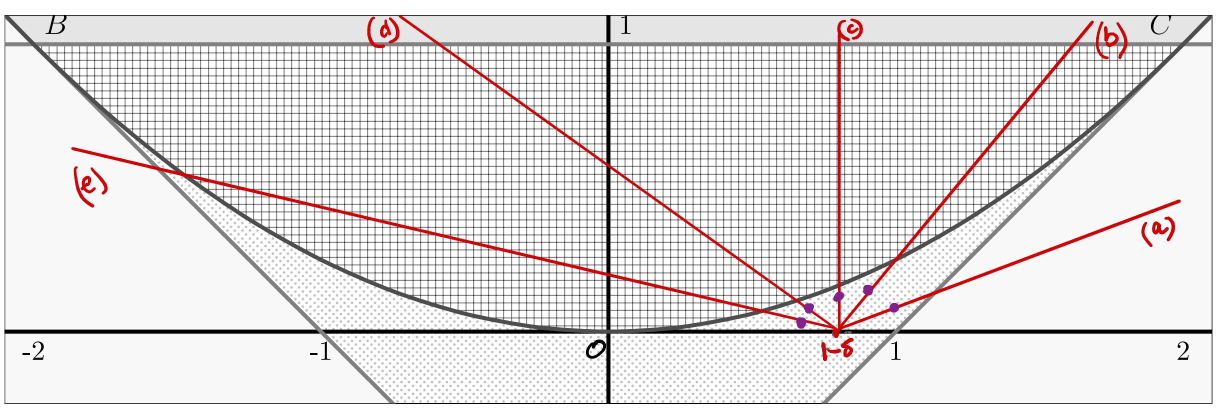
Note that (11) is the equation of a line in (T, D)-space (where the line is vertical at $T = 1-\delta$ for the case where $\alpha_1 = \alpha_2$). When ρ varies, (T, D) moves along the line (11), and under Assumption A3, equations (9) and (10) imply that this movement is continuous in ρ, and is such that D strictly increases with ρ. This allows for a simple graphical characterization of the impact of ρ on the location of the eigenvalues, and therefore on the qualitative dynamics of the system around the SS.

Consider first the case where ρ is negative, i.e., where the decisions of others have a negative effect on one's own decision. It is clear from equations (9) and (10) that

$$\lim_{\rho \to -\infty} (T, D) = E_{-\infty} \equiv (1-\delta, 0) \,.$$

This point is shown in Figure 5. Note that $E_{-\infty}$ corresponds to the case of one null and one positive/stable eigenvalue. Thus, for the limiting case $\rho \to -\infty$, the SS is locally stable and exhibits monotonic dynamics (no cycles).

Figure 5: Bifurcations



This figure shows the plane (T, D), where T is the trace and D the determinant of matrix M. The purple dots correspond to possible values of (T, D) for the model without interactions ($\rho = 0$).

Depending on the values of α_1 and α_2, there are a number of possible configurations for E_0, some examples of which are illustrated by the purple dots in Figure 5. Starting from such a point, as ρ decreases, (T, D) moves continuously along the line (11) (e.g., one of the lines denoted by (a) to (e)), starting from E_0 and approaching $E_{-\infty}$ as $\rho \to -\infty$. This leads to the following proposition.

Proposition 4. *Under Assumptions A1 and A2, as ρ decreases from 0 to $-\infty$:*

(i) Both eigenvalues of M are always inside the complex unit circle, so that the system remains locally stable.

(ii) The eigenvalues are never negative.

(iii) If either $\alpha_2 \leq 1 - \delta$, $\alpha_2 \leq \alpha_1$, or $\alpha_1 \leq 0$, then the eigenvalues are always real and positive, so that there are never cycles.

(iv) If all of the conditions from part (iii) fail to hold, then there are values ρ_1, ρ_2 with $-\infty < \rho_1 < \rho_2 < 0$ such that the eigenvalues are complex if $\rho \in (\rho_1, \rho_2)$, and real and positive for all other values of $\rho \leq 0$.

Proposition 4 indicates that, when the actions of others play the role of strategic substitutes with one's own action, this favors stability of the system. Strategic substitutability is typical of Walrasian settings, where the price mechanism mediates all strategic interactions, which is one reason why dynamic Walrasian environments are typically stable.

We now turn to exploring how the presence of strategic complementarities affects the dynamics of the system. Note first that $D \to +\infty$ when $\rho \uparrow 1$. Thus, as ρ increases from 0 toward 1, (T, D) will move continuously upward along line (11) starting from E_0, and will necessarily exit the triangle $\widehat{ABC}$ at some value $\rho < 1$. Examples of such movements are shown in Figure 5, where starting from one of the purple dots (which indicate possible values of E_0), (T, D) traverses one of the lines (a) to (e) in an upward direction. As (T, D) crosses the perimeter of triangle $\widehat{ABC}$, the SS becomes unstable. In the case of half-line (a), a real eigenvalue increases through 1 as (T, D) crosses the right edge of the triangle. In the case of half-lines (b), (c) and (d), the eigenvalues will first become complex and conjugate while still stable, and then exit the unit circle together as (T, D) crosses the top edge of the triangle. Finally, in the case of half-line (e), a real eigenvalue decreases through -1 as (T, D) crosses the left edge of the triangle.[6]

[6]Note that, for cases like (e) where (T, D) departs triangle $\widehat{ABC}$ through the left edge, there will always be an intermediate range of ρ where the eigenvalues are temporarily complex/stable, but transition back to being real/stable before stability is lost. For cases where (T, D) departs triangle $\widehat{ABC}$ through the right edge, there may or may not be such an intermediate range of complex/stable eigenvalues.

This shows that the presence of weak complementarities can radically change the dynamics of the system. In particular, if ρ is large enough (but smaller than one), it will cause the system to be unstable, even though the system would be stable in the absence of complementarity. Intuitively, the instability arises here because agents have an incentive to bunch their actions together. As a result, any initial individual desire to choose a high/low effort level because the date-t state vector (X_t, e_{t-1}) is different from the SS is then amplified in equilibrium. If the amplification is strong enough (ρ sufficiently large), e_t will be such that the resulting date-$(t+1)$ state vector (X_{t+1}, e_t) will be even more extreme than (X_t, e_{t-1}); i.e., the system will be unstable.

1.4 Bifurcations

A change of the local stability properties when a parameter varies, such as the ones described above for the case where ρ increases toward 1, is referred to as a bifurcation in the theory of dynamical systems. For a system with the structure we consider here, there are essentially three possible types of bifurcations that can occur as ρ changes. The first, known as a flip bifurcation, occurs as an eigenvalue passes through -1. The second, known as a Hopf bifurcation (more accurately referred to as a Neimark-Sacker bifurcation in discrete-time systems), occurs when a pair of complex conjugate eigenvalues crosses the unit circle together. The last, known as a fold bifurcation, occurs as an eigenvalue passes through $+1$.[7] We have already established that one of these bifurcations must occur as ρ increases toward 1. We now formally state the conditions that determine which type of bifurcation will occur.

Proposition 5. *Under Assumptions A1 and A2, as ρ increases from zero towards one, the dynamic system given by (5) will at some point become unstable and experience*

(i) a flip bifurcation if $\alpha_2 < \frac{\alpha_1}{(2-\delta)^2}$,

(ii) a Hopf bifurcation if $\frac{\alpha_1}{(2-\delta)^2} < \alpha_2 < \frac{\alpha_1}{\delta^2}$, or

(iii) a fold bifurcation if $\alpha_2 > \frac{\alpha_1}{\delta^2}$.

[7] We use the term "fold bifurcation" throughout for the case of an eigenvalue passing through $+1$ since this is the most generic case. Technically, however, depending on the precise nature of the non-linearities, other types of $+1$ bifurcations (e.g., pitchfork) are also possible. The differences between these $+1$ bifurcations will not be important for our purposes.

The conditions in Proposition 5 can be understood as follows. Recall that α_2 parameterizes the degree of sluggishness, while α_1 parameterizes the degree of decreasing returns to capital (or increasing returns when $\alpha_1 < 0$). When α_2 is sufficiently large relative to α_1, the sluggishness channel dominates, and therefore the "momentum" generated by the dynamics is positive, in the sense that a higher e_t favors a higher e_{t+1}, which in turn favors a higher e_{t+2}, etc. As a result, instability in this case emerges via a positive eigenvalue, i.e., a fold bifurcation.

On the other hand, when α_2 is sufficiently small relative to α_1, the decreasing returns channel dominates, causing "negative" momentum: a higher e_t favors a *lower* e_{t+1}, which in turn favors a *higher* e_{t+2}, etc. This produces "jagged" oscillations in e, wherein e flips back and forth over e^s each period. In this case, instability emerges via a negative eigenvalue, i.e., a flip bifurcation.

Finally, if α_2 is of an intermediate magnitude relative to α_1, the sluggishness and decreasing returns channels play off each other, causing more complicated dynamics. Specifically, the sluggishness channel dominates for a while, which favors a sustained rise or fall in e, until the accumulated effect of this rise or fall (as reflected in X) becomes extreme enough that the decreasing returns channel takes over, causing e to then reverse direction. The end result is "smooth" oscillations in e, wherein e may spend more than one consecutive period on one side of e^s before switching to the other. In this case, instability emerges via a pair of complex eigenvalues, i.e., a Hopf bifurcation.

Letting $\bar{\rho} \in (0, 1)$ denote the value of ρ at which the bifurcation occurs, for ρ in a neighborhood of $\bar{\rho}$ the qualitative dynamics of the linearized system are straightforward to characterize: for $\rho < \bar{\rho}$, the system converges to the no-interaction SS, SS_0, as $t \to 0$, while for $\rho > \bar{\rho}$ the system becomes unbounded as $t \to 0$. Convergence or divergence happens either monotonically (fold bifurcation), with smooth oscillations (Hopf bifurcation), or with jagged oscillations (flip bifurcation).

When considering the full non-linear version of the model, however, the local qualitative dynamics near the bifurcation point $\bar{\rho}$ will generally be more complicated and will depend on the nature of the non-linearities embodied in the function F. The following proposition establishes some key local properties of the non-linear dynamic system near the bifurcation point.

Proposition 6. *Under Assumptions A1 and A2, as ρ increases through $\bar{\rho}$:*

(i) If the system experiences a flip bifurcation, then the steady state remains unique, and a 2-period cycle[8] either emerges or disappears.

(ii) If the system experiences a Hopf bifurcation, then the steady state remains unique, and an additional isolated closed invariant curve[9] either emerges or disappears.

(iii) If the system experiences a fold bifurcation, then two additional steady states appear, one with $e < e^s$ and another with $e > e^s$.

Several points are worth emphasizing/clarifying about Proposition 6. First, note that extra SSs appear only in the fold bifurcation case. Second, the isolated closed invariant curve referred to in the Hopf case can be visualized as a closed loop surrounding SS_0 in (X, e)-space. Third, if the system begins on the 2-cycle in the flip case or on the closed invariant curve in the Hopf case, then the system will remain on that cycle/curve—and therefore experience continual fluctuations—indefinitely. Fourth, while the cycle in the flip case is necessarily two periods long, the Hopf case can in principle be associated with cycles of any length longer than two. Thus, we view the latter case as more likely to be relevant for studying macroeconomic fluctuations.

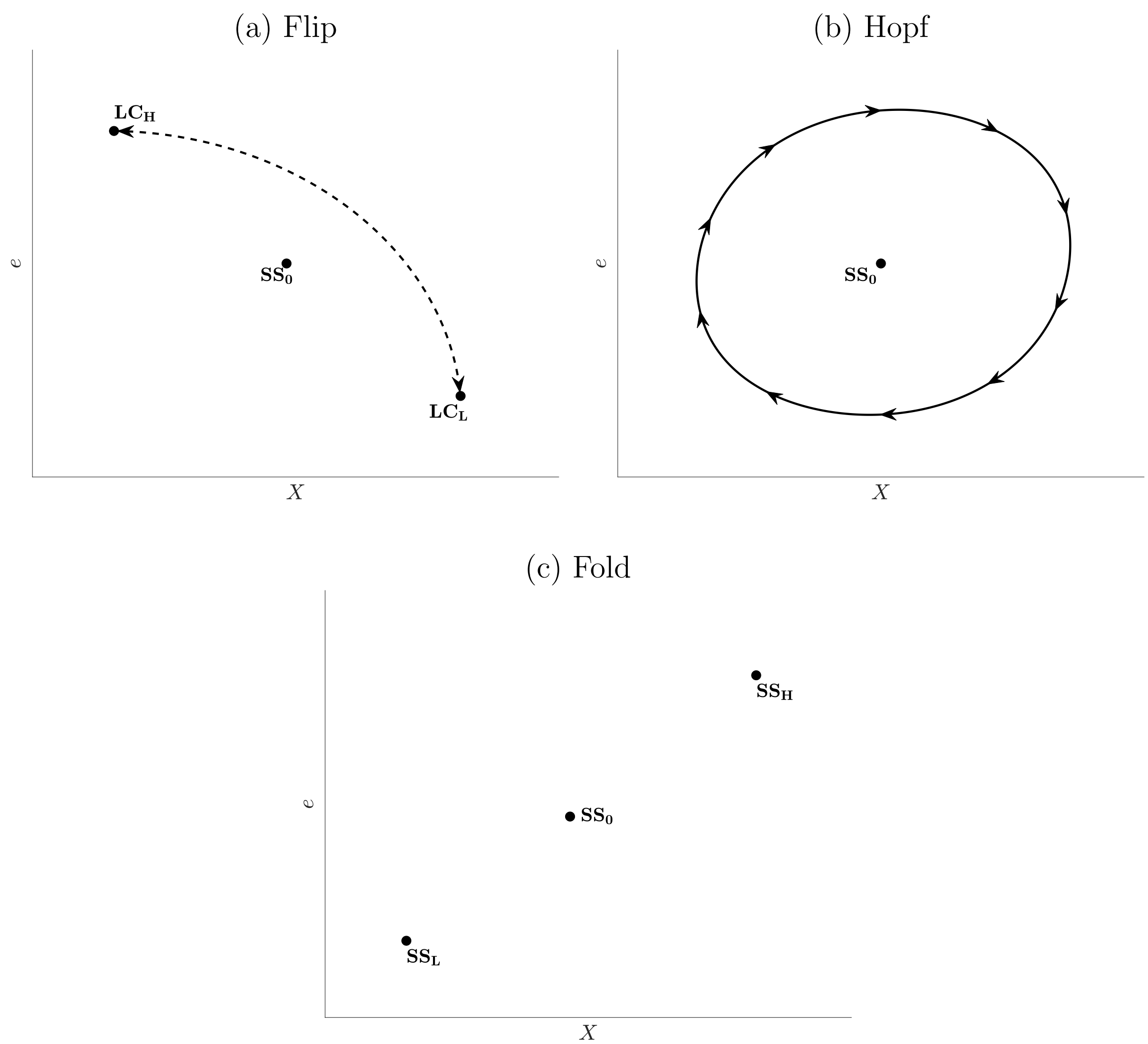
In each case, the bifurcation is associated with a qualitative change in the global dynamics of the system, as captured by the emergence or disappearance of a special set of points: the 2-cycle in the flip case, the closed invariant curve in the Hopf case, and the two additional SSs in the fold case. These sets are illustrated in Figure 6. In the flip case shown in panel (a), the 2-cycle is indicated by the points LC_H and LC_L. Starting from any point on this 2-cycle, the system jumps back and forth each period, exhibiting jagged oscillations. Since

[8] Letting $x \equiv (X, e)$, and writing the evolution of the equilibrium system as $x_t = G(x_{t-1})$, a k-period cycle is defined as a fixed point of the system $x' = G^k(x)$ that is not also a fixed point of the system $x' = G^j(x)$ for some $j = 1, \ldots, k-1$. Here, G^k denotes the k-th iterate of G, i.e., $G^1(x) = G(x)$, $G^k = G(G^{k-1}(x))$ for $k \geq 2$.

[9] An invariant set of the system $x_t = G(x_{t-1})$ is a set of points A such that $x \in A$ implies $G(x) \in A$. A closed invariant curve is an invariant set that is closed and can be represented as the image of a continuous function defined on an interval of $\mathbb{R}$. A closed invariant curve A is isolated if trajectories beginning sufficiently near (but not in) A are not themselves closed.

the system is non-linear, the 2-cycle in this case is also generically a limit cycle, meaning that nearby trajectories either converge to or diverge from it over time.[10]

Figure 6: Emerging Sets



Notes: In the flip case (panel (a)), a 2-cycle exists near the bifurcation. In the Hopf case (panel (b)), a closed invariant loop surrounding SS_0 exists near the bifurcation. In the fold case (panel (c)), two additional SSs appear for $\rho > \bar{\rho}$.

In the Hopf case shown in panel (b) of Figure 6, starting from any point on the closed loop surrounding SS_0, the system traverses it indefinitely, exhibiting smooth cycles (i.e., it takes more than two periods for the system to complete a full revolution around the loop).

[10] An example of a 2-cycle that is not also a limit cycle can be found in the linear univariate system $x_{t+1} = -x_t$. In this system, $\pm x$ is a 2-cycle for any $x > 0$, but trajectories beginning near any such cycle neither converge to nor diverge from it.

Again, since the system is non-linear, trajectories beginning near the loop generically either converge to or diverge from it over time. Thus, as with the 2-cycle in the flip case, we will refer to this loop as a limit cycle.[11]

Finally, in the fold case in panel (c), we simply have two additional steady states, a "low" one $SS_L \equiv (X_L, e_L)$, and a "high" one $SS_H \equiv (X_H, e_H)$.

Flip and Hopf bifurcations can be classified as either supercritical or subcritical. In the supercritical case, the limit cycle emerges just after ρ increases past $\bar{\rho}$, and is attractive, in the sense that all nearby trajectories—including those that begin anywhere in between the SS and the limit cycle—converge to it over time. In the subcritical case, the limit cycle exists on the "stable" side of the bifurcation (i.e., for $\rho < \bar{\rho}$), and is repellent, in the sense that nearby trajectories diverge from it (with those that begin between the SS and the limit cycle converging to the still-stable SS). In this case, the limit cycle will disappear as ρ increases through $\bar{\rho}$. The following Proposition establishes that the flip and Hopf bifurcations are always supercritical in this model.

Proposition 7. *Suppose Assumptions A1 and A2 hold. Then the flip and Hopf bifurcations are supercritical.*

Next, in the fold bifurcation case, it was shown in Proposition 6 that, as the no-interaction SS loses stability, two additional SSs emerge. The following proposition establishes that these two new SSs are each locally stable.

Proposition 8. *In the case of the fold bifurcation, the two other steady states SS_L and SS_H are locally stable, so that the system displays hysteresis.*

As noted earlier, regardless of the type of bifurcation, when the no-interaction SS, SS_0, becomes unstable, it does so because the complementarities—which cause agents to want to bunch their actions together—have become strong enough locally that any initial desire by agents to deviate from the SS action e^s gets successively larger over time through the

[11] Since time is discrete, the steps taken as the system moves around the continuous loop are also discrete. Thus, starting from any point on the loop, despite the fact that the system will remain on that loop forever, it (a) need never return to its exact starting point, and (b) cannot possibly visit every point on the loop. For these reasons, the loop is not technically a cycle. However, since for our purposes the distinction is mainly a technical one, we will nonetheless refer to the loop as a limit cycle.

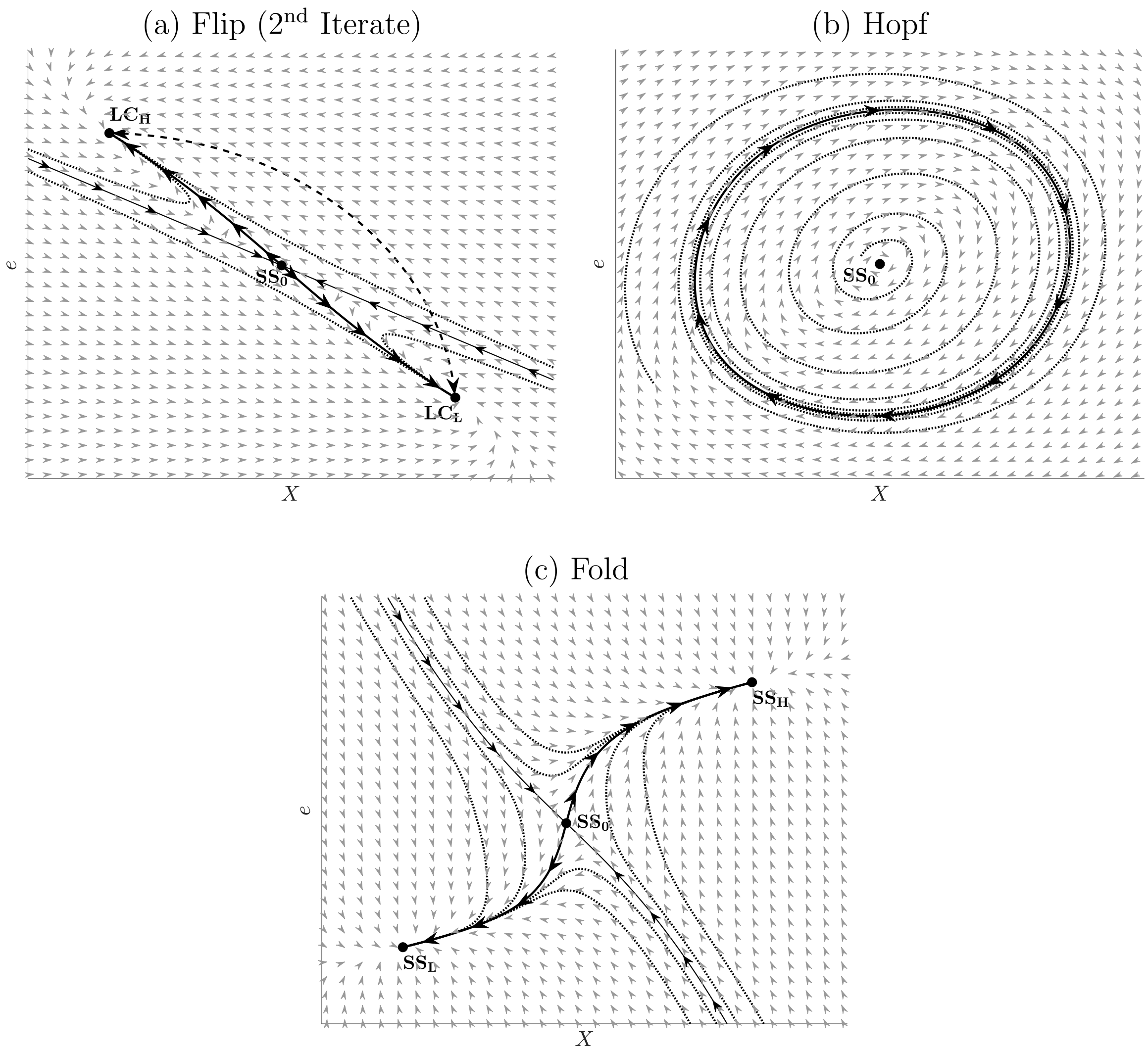
dynamic mechanisms of accumulation and sluggishness. In the linearized system this is the end of the story. However, once we introduce non-linearities into the interaction function F, this is no longer the case. In particular, because the slope of F becomes flatter as e moves away from e^s in either direction, the complementarities become weaker as the system diverges from SS_0. Thus, as the dynamics push the economy further and further from the unstable SS_0, the complementarity mechanism that was responsible for that instability eventually weakens to the point that the system no longer favors divergence. Put differently, the weaker complementarities away from SS_0 reduce the bunching incentive enough so that the private incentives—i.e., those present in the no-interactions model of Section 1.1, which favor stability—take over and contain the system, preventing it from exploding.

As a result of this structure, just as SS_0 becomes unstable, one or more additional stable sets appear. In the flip case, the additional stable set is the 2-cycle referred to in Proposition 6(i). Instability arises locally in this case in the form of ever-growing jagged oscillations in e around its unique SS value e^s. As $|e - e^s|$ continues to grow, however, the complementarities eventually fade sufficiently that e stops moving further away from e^s, while continuing to oscillate in a jagged away; i.e., the system approaches the (attractive) 2-cycle. In the Hopf case, the intuition is essentially the same, except the additional stable set is the closed invariant curve referred to in Proposition 6(ii), and the oscillations in question are smooth rather than jagged.

The post-bifurcation dynamics for the flip and Hopf cases are illustrated by the phase diagrams in panels (a) and (b) of Figure 7. The light gray arrows in all cases indicate the local direction of motion, with examples of typical trajectories given by the dotted curves. In the Hopf case in panel (b), we see that, regardless of whether they begin inside or outside the limit cycle, as long as the system does not start exactly at SS_0 the trajectories always converge to it over time.

Because of the violent nature of the oscillations in the flip case, a standard phase diagram for that system would be difficult to make sense of. Letting $x_t \equiv (e_t, X_t)$ and writing our system as $x_t = G(x_{t-1})$, in panel (a) we instead present the phase diagram for the 2^{nd} iterate of the system, i.e., for the map $G^2(x) \equiv G(G(x))$ that gives the position of the system two periods later when the current system is at x. The dynamics illustrated in the figure can

Figure 7: Global Dynamics After the Bifurcation



Notes: This figure shows the paths of trajectories when starting away from a SS.

be understood as follows. First, note that the two points LC_H and LC_L of the 2-cycle are both stable SSs of G^2. Thus, trajectories of G^2 that begin near one of the 2-cycle points will converge to it over time. In terms of the original map G, this convergence corresponds to jagged oscillations in both e and X, each typically jumping back and forth across its SS_0 value each period, with the system converging eventually to the 2-cycle. Second, because SS_0 is actually a saddle in this case (i.e., it has one stable and one unstable eigenvalue), it has a stable arm (i.e., a saddle path) and an unstable arm. The saddle path is indicated by the thin solid curve with arrows pointing toward SS_0, indicating that if the system begins exactly on this path it will eventually converge to SS_0. The unstable arm, meanwhile, is indicated by the thick solid curve with arrows pointing away from SS_0. Examples of typical trajectories for the global system are again given by dotted curves. As the figure shows, all trajectories of G^2 that do not begin either at SS_0 or on the saddle path will converge to one of the 2-cycle points. This in turn implies that all such trajectories of the original map G will converge to the 2-cycle in the manner described above.

The fold case differs from the flip and Hopf cases in that instability does not arise in the form of oscillations, but in the form of monotonic divergence from SS_0. As a result, the additional stable sets that emerge as the system loses stability—i.e., the two new SSs referred to in Proposition 6(iii)—do not feature oscillations either. The dynamics for this case are illustrated in panel (c) of Figure 7. As in the flip case, SS_0 is a saddle, and therefore possesses a saddle path (thin solid curve with arrows pointing towards SS_0) and an unstable arm (thick solid curve with arrows pointing away from SS_0). If the system begins exactly on the saddle path, it will converge to SS_0. Trajectories beginning anywhere in the region northeast of this saddle path will converge to SS_H, while those beginning anywhere southwest of it will converge to SS_L. In this way, we see that a small perturbation from SS_0 can permanently affect the system, i.e., the dynamics display hysteresis.

As a final point, recall that a necessary condition for a fold bifurcation is $\alpha_2 > \alpha_1/\delta^2$. Given that, quantitatively, δ is typically taken to be quite small (e.g., on the order of 0.05 or less for a quarterly model), this is only possible for some $\alpha_2 < 1$ if capital exhibits increasing returns ($\alpha_1 < 0$) or at most very weak decreasing returns ($\alpha_1 \in (0, \delta^2)$). Thus, the general insight we take away is that limit cycles (whether smooth or jagged) are likely to emerge in our setting if capital exhibits even modest decreasing returns, and if complementarities are (i) strong enough to create instability near the SS, but (ii) tend to die out as the economy moves away from the SS. In an economic environment, it is quite reasonable to expect that complementarities are likely to die out if activity gets very large. For example, if investment demand gets sufficiently large, some resource constraints are likely to become binding, causing incentives akin to strategic substitutes to emerge instead of complementarities. Similarly, physical constraints, such as non-negativity restrictions on investment or capital, would prevent the economic system from diverging to zero. Such forces will generally favor the emergence of attractive limit cycles in the presence of complementarities.

1.5 Summary

The take-away of that section is the following. First, substitutabilities do not create fluctuations. Second, weak complementarities are always enough for the emergence of a bifurcation; *(i)* when $\alpha_1 < 0$, only hysteresis-inducing fold bifurcation; when $\alpha_1 > 0$, we have flip bifurcations for low α_2, Hopf bifurcations for intermediate α_2 and hysteresis-inducing fold for high α_2; *(iii)* limit cycles (smooth or 2-cycles) are attractive, and *(iv)* when there is hysteresis, the two emerging new steady states are locally stable.

2 Adding Forward-Looking Elements

In the previous section, we illustrated how agent interaction can create limit cycles when agents are accumulating goods and when there is inertia in their behavior. However, the framework we explored purposely omitted any forward-looking elements on the part of agents. This omission has the advantage of allowing us to highlight key forces that can give rise to limit cycles, without needing to simultaneously address issues regarding the extra potential source of multiple equilibria that may arise when agents are forward-looking. In this section, we extend the results of the previous section to the case where individual agents' decision rules take the form

$$e_{it} = \max\left\{\alpha_0 - \alpha_1 X_{it} + \alpha_2 e_{it-1} + \alpha_3 \mathbb{E}_{it}\left[e_{it+1}\right] + \mathbb{E}_{it}\left[F\left(e_t\right)\right], 0\right\}\ , \tag{12}$$

with $\alpha_0 > 0$, $-1 < \alpha_1 < 1$, $0 < \alpha_2 < 1$, and $0 < \alpha_3 < 1$. As before, α_1 controls degree of decreasing returns to capital, while α_2 controls the degree of sluggishness. The new parameter α_3, meanwhile, determines the degree of "forward-looking-ness". The evolution of X continues to obey (1), and the function $F(\cdot)$ again controls the type and degree of strategic interactions between agents' decisions. Also as before, we will typically ignore the non-negativity constraint embodied in the max operator, and focus on symmetric equilibria. We also omit any exogenous stochastic driving forces, so that we can remove the expectation operators in equation (12). In symmetric equilibrium, we then obtain the system

$$\begin{aligned} e_t - F(e_t) &= \alpha_0 - \alpha_1 X_t + \alpha_2 e_{t-1} + \alpha_3 e_{t+1}\,, \\ X_{t+1} &= (1-\delta) X_t + e_t\,. \end{aligned} \tag{13}$$

Environments with forward-looking agents typically also require, at a minimum, the imposition of some kind of transversality condition (TVC) in order to pin down a unique equilibrium path given a particular initial state. Depending on the precise structure of the model, this TVC is typically derived from agents' optimization problem, along with possibly a no-Ponzi/no-bubbles condition. In solving models like ours that do not feature any sources of permanent growth, the TVC is typically used to rule out solution paths that either become unbounded or involve one or more variables converging to zero. Since we have not explicitly formulated agents' optimization problems, we impose the following version of a TVC, which is a simple way to directly impose the above features for our model:

$$\liminf_{t\to\infty} e_t > 0\,, \qquad \limsup_{t\to\infty} e_t < \infty\,. \tag{TVC}$$

With this definition, a solution to the model given an initial state (e_{-1}, X_0) is defined as a sequence $\{(e_t, X_{t+1})\}_{t=0}^{\infty}$ satisfying the difference equations (13) along with (TVC).

The first-order approximation of the equilibrium of the system (13) around a particular SS given by $SS^* \equiv (e^*, X^*)$ is given by (omitting constant terms)

$$\begin{pmatrix} X_{t+1} \\ e_{t+1} \\ e_t \end{pmatrix} = \underbrace{\begin{pmatrix} 1-\delta & 1 & 0 \\ \frac{\alpha_1}{\alpha_3} & \frac{1-F'(e^*)}{\alpha_3} & -\frac{\alpha_2}{\alpha_3} \\ 0 & 1 & 0 \end{pmatrix}}_{M} \begin{pmatrix} X_t \\ e_t \\ e_{t-1} \end{pmatrix}. \tag{14}$$

Since M is a 3×3 matrix, it has at least one real eigenvalue. Let λ_2 denote the largest real eigenvalue of M, and $\lambda_{11}, \lambda_{12}$ the remaining two eigenvalues, with $\lambda_{11} \leq \lambda_{12}$ when they are real, and λ_{11} the eigenvalue with negative imaginary part when they are complex. When the solution is determinate—i.e., when there is a unique path satisfying (13) and (TVC)—the qualitative dynamics near SS^* will be dictated by two of these three eigenvalues. In particular, positive eigenvalues will be associated with monotonic dynamics, complex eigenvalues with smooth oscillations, and negative eigenvalues with jagged oscillations (e.g., period-2 cycles). If exactly two of the three eigenvalues are stable (i.e., inside the complex unit circle), then the unique solution will locally feature convergence to SS^*. If all three of the eigenvalues are stable then the solution will be indeterminate. Finally, if none or only one of the three eigenvalues are stable, then the linearized system (14) has no solution, though this does not necessarily imply that the original *non-linear* system has no solution. Indeed, as we shall see, there may in some cases be solutions featuring limit cycles around this or another SS, or convergence to some other SS.

2.1 No-Interaction Case

As in the backward-looking case, it is useful to first characterize the no-interaction version of the model, i.e., where $F(e) = 0$ for all e. We make the following assumption, which extends A1 to the forward-looking model.

Assumption A4. *The parameters satisfy:*

(i) $\alpha_1 > -(1 - \alpha_2 - \alpha_3)\delta$,

(ii) $\alpha_1 < \left(\sqrt{\alpha_2} - \sqrt{1 - \delta}\right)^2$.

Letting $\rho^* \equiv 1 - \alpha_2 - \alpha_3 + \alpha_1/\delta$, Assumption A4(i) implies that $\rho^* > 0$. The unique no-interaction SS is then given by $SS_0 = (e^s, X^s)$, where $e^s = \alpha_0/\rho^* > 0$ and $X^s = e^s/\delta$. We then have the following proposition regarding the eigenvalues of M for $SS^* = SS_0$.

Proposition 9. *Suppose there are no interactions (i.e., $F(e) = 0$) and Assumption A4 holds. Then there exists an $\bar{\alpha}_3 \in (0, 1]$ such that the eigenvalues of M are real with $0 < \lambda_{11}, \lambda_{12} < 1 < \lambda_2$ if and only if $\alpha_3 \leq \bar{\alpha}_3$.*

Since, as in the backward-looking case, we are mainly interested in situations where, in the absence of interactions, agents' behavior favors (determinate) convergence without oscillations, we will typically make the following additional assumption.

Assumption A5. *$\alpha_3 \leq \bar{\alpha}_3$, where $\bar{\alpha}_3$ is defined in Proposition 9.*

Under Assumptions A4 and A5, Proposition 9 implies that, for the no-interaction case, the solution is determinate, with solutions paths converging to the unique SS (SS_0) monotonically.

2.2 Adding Interactions

2.2.1 Period-t Equilibrium and Steady State(s)

Returning to the case with interactions, we again impose Assumption A2, which parameterizes the shape of the interaction function F by its slope ρ at $e = e^s$. We will also maintain Assumption A3 which imposes that complementarities, if they exist, are weak, in the sense that $F'(e) < 1$. The following proposition extends Propositions 2 and 3 to the forward-looking case. Note that we use the term "period-t equilibrium" to mean a value of e_t that satisfies the first equation in (13) for a given current state (e_{t-1}, X_t) and anticipated value of e_{t+1}.

Proposition 10. *Suppose Assumptions A2-A4 hold. Then there exists a unique period-t equilibrium, and:*

(i) If $\rho \leq \rho^*$—which is necessarily the case if either $\rho \leq 0$ (substitutabilities) or $\alpha_1 \geq \delta(\alpha_2 + \alpha_3)$—then SS_0 is the unique steady state.

(ii) If $\rho > \rho^*$—which is only possible if both $\rho > 0$ (complementarities) and $\alpha_1 < \delta(\alpha_2 + \alpha_3)$—then there are two steady states in addition to SS_0 (one with $e > e^s$ and one with $e < e^s$).

2.2.2 Bifurcations

We now consider how the economy's dynamics are affected when the type and degree of strategic interaction—as parameterized by ρ—is allowed to vary. As established in Proposition 9, when $\rho = 0$, two of the eigenvalues of M—λ_{11} and λ_{12}—are real, positive and stable, while the third eigenvalue—λ_2—is real, positive and unstable. Thus, beginning from $\rho = 0$, the system has a saddle-path stable configuration with solution paths that converge monotonically to SS_0. If this dynamic system undergoes one or more bifurcations as ρ moves away from zero, the first bifurcation[12] must be of one of the following four types:[13]

(i) λ_2 decreases through 1 (indeterminacy-inducing fold).

(ii) λ_{12} increases above 1 (hysteresis-inducing fold).

(iii) λ_{11} decreases through -1 (flip).

(iv) λ_{11} and λ_{12} first become complex and then leave the unit circle together (Hopf).

It is worth briefly clarifying what these bifurcations generally look like in our model. Type (i) corresponds to the case where the system becomes locally indeterminate—in the sense of generating a continuum of equilibrium paths satisfying (TVC)—and two additional unstable SSs emerge (i.e., the matrix M corresponding to each new SS has three unstable eigenvalues). This type of bifurcation, which we refer to as an indeterminacy-inducing fold, is the one typically emphasized in the sunspot literature (see, for example, Benhabib and Farmer [1994]).

The type-(ii) bifurcation corresponds to the case where, outside of a measure-zero subset of the initial state space, the linearized system (14) has no paths satisfying (TVC), and two additional locally stable and determinate SSs (i.e., the M corresponding to each new SS will have two stable eigenvalues and one stable one) of (13) will emerge. We refer to this as a hysteresis-inducing fold bifurcation, since small perturbations from SS_0 will typically move the system to a different SS.

Finally, the type-(iii) and type-(iv) bifurcations correspond to the emergence of jagged and smooth limit cycles, respectively. We discuss the nature of these bifurcations further below. The following propositions characterize the conditions under which each of the above bifurcations will occur as ρ moves away from zero.

Proposition 11. *Suppose Assumptions A2-A5 hold. Then as ρ decreases from zero to $-\infty$, system* (13) *does not experience a bifurcation. Thus, the solution continues to be determinate, with solution paths converging to SS_0.*

[12] We focus only on the first bifurcation that the system undergoes, i.e., the first time an eigenvalue crosses the complex unit circle as ρ increases or decreases from zero. In general, there may be further bifurcations beyond this point.

[13] This follows from the fact that the coefficients of the characteristic polynomial of M (see equation (A.16) in Appendix A.2) are continuous in ρ, and therefore so are the roots of that polynomial.

Proposition 12. *Suppose Assumptions A2-A5 hold, and that, as ρ increases from zero towards one, a bifurcation occurs at some point. We have the following:*

(i) If $\alpha_2 < \frac{\alpha_1}{(2-\delta)^2} + \alpha_3$, then the first bifurcation is a flip.

(ii) If $\frac{\alpha_1}{(2-\delta)^2} + \alpha_3 < \alpha_2 < \frac{\alpha_1}{\delta^2} + \alpha_3$, then the first bifurcation is an indeterminacy-inducing fold if $\alpha_3 > \alpha_2(1-\delta)$, and a Hopf if $\alpha_3 < \alpha_2(1-\delta)$.

(iii) If $\alpha_2 > \frac{\alpha_1}{\delta^2} + \alpha_3$ (which is necessarily the case if $\alpha_1 < 0$), then the first bifurcation is a hysteresis-inducing fold.

(iv) The two types of fold bifurcation are only possible if $\alpha_1 < \delta(\alpha_2 + \alpha_3)$.

Several things are worth noting about Proposition 12. First, although we have extended our framework to include forward-looking behavior, we find that limit cycles (both smooth and jagged) are still a possibility. Unlike in the backward-looking case, indeterminacy-inducing fold bifurcations are also now a possibility.

Second, the three ranges for α_2 in parts (i)-(iii) are, for the three types of bifurcation that are possible in the backward-looking model (flip, Hopf, hysteresis-inducing fold), directly comparable to the conditions shown in Proposition 5 for the corresponding case (with similar intuition), and in particular the bounds of the regions each differ only by the extra term $+\alpha_3$.

Third, note that indeterminacy crucially relies on the coordination of agents' expectations about the future, and is therefore more likely to occur when agents are highly forward-looking, i.e., α_3 is large. This can be seen directly in part (ii) of Proposition 12, where α_3 must be sufficiently large in order to have an indeterminacy-inducing fold as the first bifurcation.

Fourth, as long as capital exhibits decreasing returns ($\alpha_1 > 0$), then if δ is small neither type of fold bifurcation is likely to occur (i.e., the condition in part (iv) is unlikely to hold).

Finally, note that Proposition 12 does not imply that, when agents are forward-looking, a bifurcation will always occur as ρ increases from zero to one. This is in contrast to the backward-looking case, where local instability always emerged for some $\rho \in (0, 1)$.

Let us emphasize that the nature of a Hopf bifurcation in the presence of forward-looking behavior is somewhat different from the case without it.[14] For the forward-looking case where the first bifurcation is a Hopf, as the two complex stable eigenvalues become unstable, depending on the nature of the non-linearities the system will in many cases continue to exhibit a structure analogous to the saddle-path one that existed just prior to the bifurcation. Specifically, even if an attractive one-dimensional limit cycle appears, it will generally only be attractive on a two-dimensional manifold—which we refer to as the non-explosive manifold—in our three-dimensional phase space.[15] The transversality condition (TVC) will then force e_t to jump—for any given initial values of X_t and e_{t-1}—onto this manifold. Figure 8 presents a phase diagram that illustrates this configuration, which we refer to as a saddle limit cycle. The light grey manifold in the figure contains an attractive limit cycle, and beginning from any point initially on this manifold (except the SS), the system converges to the cycle (dark grey lines). The dynamics of the system beginning from any other point, however, are dominated by the remaining (real) unstable eigenvalue, causing paths emanating from such points to explode (black arrows).[16]

2.3 Numerical Illustration

In this section, we graphically illustrate the results obtained in Proposition 12. In Figure 9, each panel shows, for a given δ and α_1, the space $(\alpha_2, \alpha_3) \in [0, 1]^2$. Up to the fineness of the grid, we therefore spans the entire parameter space for (α_2, α_3). Each row corresponds to a value of δ (.025, .4 and .9) and each column to a value of α_1 (-.1,0,.1 and .2). The white space corresponds to the parameters configurations for which the no-interaction model is not determinate and a-cyclical with a unique steady state (Assumptions A4 and A5). In each parameters configuration, we increase complementarity parameter ρ up to the first bifurcation, it it exists, and report with a different colour the type of bifurcation that occurs.

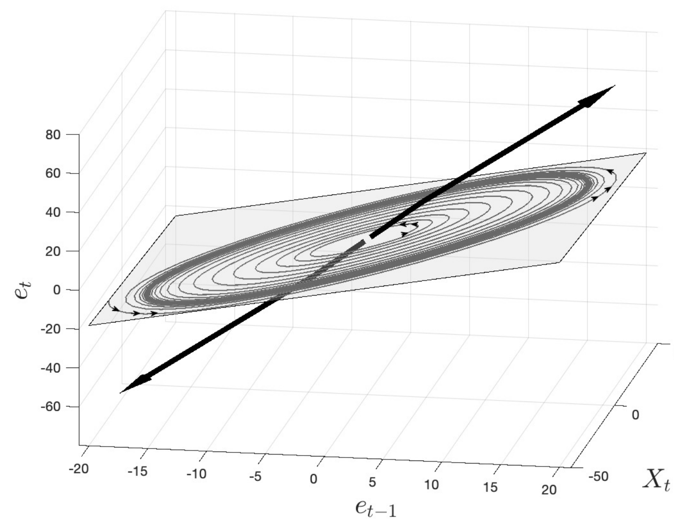
First row corresponds to $\delta = 0.025$. Going from left to right, we increase the value of α_1.

[14] A similar property holds for the flip bifurcation.

[15] Formally, if A is the eigenspace associated with the two complex eigenvalues, then the non-explosive manifold is the invariant analytic two-dimensional manifold that is tangent to A at the SS.

[16] The dynamic system we are considering is three-dimensional, which allows for the possibility of such a saddle limit cycle to appear via a Hopf bifurcation. In a two-dimensional system, on the other hand, an attractive limit cycle that appears via a Hopf bifurcation would necessarily be associated with indeterminacy, since beginning from any point in the phase space the system would converge to the limit cycle, and therefore the transversality condition would be satisfied. This latter case has appeared often in the macroeconomic literature. In contrast, the type of configuration implied by Figure 8 is less common.

Figure 8: A Saddle Limit Cycle



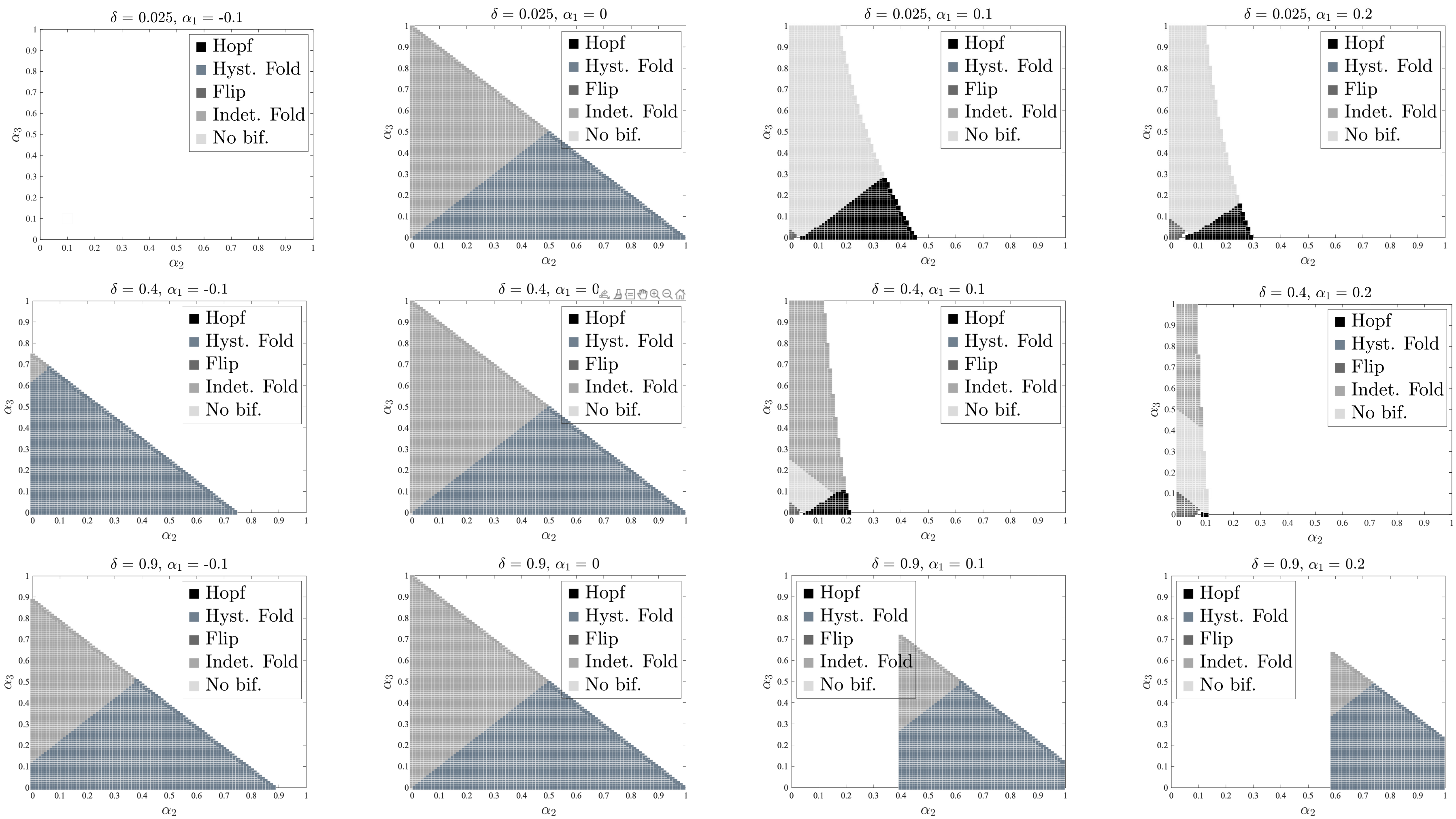
Notes: The light grey zone is the non-explosive manifold (which is here locally drawn as a linear plane, but which is in general nonlinear). The dark grey lines are two paths that converge to the limit cycle, one from the inside and one from the outside. The black lines are two paths for which the jump variable has not jumped onto the non-explosive manifold, and which therefore diverge.

When α_1 is negative, there are no acyclical steady states absent of complementarities. When α_1 is zero, the first bifurcation is indeterminacy-inducing fold if there is enough "forward-looking-ness" (α_3 large enough), and hysteresis-inducing fold otherwise. When α_1 becomes positive, we will have a Hopf bifurcation (smooth cycles) if sluggishness (α_2) is large enough, no bifurcation (and dampening cycles) if "forward-looking-ness" is large enough. Note that for small α_2 and α_3, we will have a Flip bifurcation, and hence non-smooth cycles.

Second row corresponds to $\delta = 0.4$. For non negative α_1, we have either hysteresis-inducing fold bifurcation or indeterminacy-inducing fold when there is enough "forward-looking-ness". As α_1 gets positive, we will have Hopf if sluggishness is large enough, indeterminacy-inducing fold if there is enough "forward-looking-ness" and Flip bifurcation for small α_2 and α_3. Note that the zone for smooth cycles (Hopf) is reduced as compared to a smaller δ configuration.

Third row corresponds to $\delta = 0.9$. As there is little persistence though X (because depreciation is large), we need enough sluggishness to make sure that the no interaction model is acyclical when $\alpha_1 > 0$. As in the previous cases, we will have indeterminacy-inducing fold if there is enough “forward-looking-ness” and hysteresis-inducing fold otherwise.

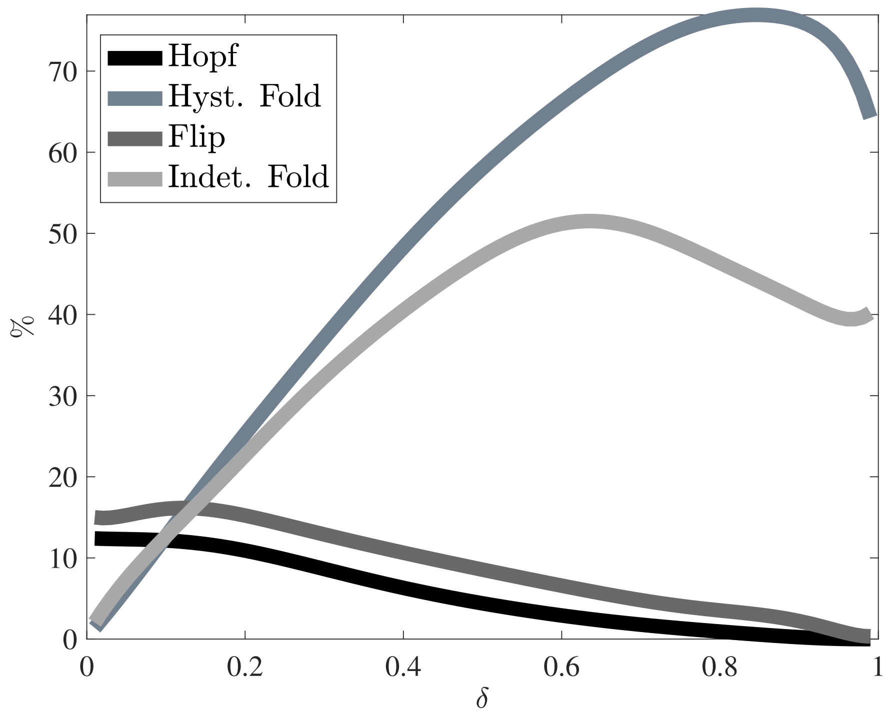
Figure 9: Nature of the Bifurcation when Complementarities Are Increased



Notes: For each δ, we look at a fine grid over $(\alpha_1, \alpha_2, \alpha_3) \in [-1, 1] \times [0, 1]^2$. For each triplet $(\alpha_1, \alpha_2, \alpha_3)$ that satisfies non-cyclical and determinate convergence of the no-interaction SS, we increase ρ from 0 to 1 and look at the nature of the first bifurcation (if there is one).

Finally, we explore the effect of the value of depreciation δ in the following way. For a given δ, we consider all the $(\alpha_1, \alpha_2, \alpha_3)$ such that the no-interaction SS is determinate with non-cyclical convergence. For each of these triplets, we determine which bifurcation is the first to occur (if any in fact do) as ρ increases from zero to one. For each δ, we may then compute the percentage of cases corresponding to each of the four possible bifurcation types. The difference between the sum of these percentages and 100% corresponds to the percentage of cases with no bifurcation. We can then compute these percentages for all values of δ. The results are reported in Figure 10. As can be seen, low values of δ favor limit cycles (flip or Hopf), while high values of δ favor indeterminacy and hysteresis.

Figure 10: Nature of the First Bifurcation When Complementarities Are Increased, as a Function of δ



Notes: For a given δ, we consider all the $(\alpha_1, \alpha_2, \alpha_3)$ such that the no-interaction SS is determinate with non-oscillating convergence. For each of these triplets, we increase ρ up to the first bifurcation (if there is one). For each δ, we may then compute the percentage of cases corresponding to each of the four possible bifurcation types. The difference between the sum of these percentages and 100% corresponds to the percentage of cases with no bifurcation. The Figure reports these percentages as a function of δ.

2.4 Summary

The take-away of this section, as obtained from the analytical necessary and sufficient conditions and from the numerical illustration, is as follows. Forward-looking behaviour does not prevent limit cycles (both smooth and jagged) to occur, but now another type of bifurcation is possible, namely indeterminacy-inducing fold bifurcation. Then, conditions on sluggishness (α_2 parameter) for the occurrence of flip, Hopf and hysteresis-inducing fold bifurcations are quite similar to their backward model counterparts. Finally, indeterminacy crucially relies on the coordination of agents' expectations about the future, and is therefore more likely to occur when agents are highly forward-looking, i.e., α_3 is large.

Conclusion

We have presented a reduced form model of accumulation and interactions between agents. The model can be seen either as an Agent-Based model or as the DSGE one, depending on how forward-looking the agents are. When complementarities between agents are increased (but stay weak in the sense that there are no multiple period-t equilibria), non trivial dynamic properties can emerge, depending on three mechanisms: the impact (positive or negative) of past accumulation on current actions, the strength of sluggishness in actions and the strength of the forward looking behaviour of economic agents. Depending on the relative strength of those three forces, hysteresis, limit cycles or sunspot equilibria can emerge.

Therefore, complementarities in actions and accumulation allow for the emergence of fluctuations in otherwise stable and non-oscillating economies. The assumption of rational expectations plays a role in determining which type of emergent phenomenon will occur, but is by no mean necessary except for indeterminacy.

We view our results as showing that minimal non-linearity and weak complementarities are enough to generate rich dynamics, and in particular endogenous fluctuations, as we have given a fully microfounded and estimated example in Beaudry, Galizia, and Portier [2020].

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Appendix

A Proofs

A.1 The Backward-Looking Model

Proposition 1

The two eigenvalues of matrix M_L are given by (see Section 1.3 for the case where $\rho = 0$)

$$\lambda, \overline{\lambda} = \frac{T}{2} \pm \sqrt{\left(\frac{T}{2}\right)^2 - D} \tag{A.1}$$

where

$$T = \alpha_2 - \alpha_1 + 1 - \delta\,, \qquad D = \alpha_2(1-\delta)\,. \tag{A.2}$$

We are interested in establishing conditions under which (T, D) is in the zone of Figure 4 corresponding to eigenvalues that are both real, positive, and stable. This zone is defined by the following four conditions:

$$D < T^2/4\,, \tag{A.3}$$
$$T > 0\,, \tag{A.4}$$
$$D > 0\,, \tag{A.5}$$
$$D > T - 1\,. \tag{A.6}$$

Condition (A.3) is the condition for real eigenvalues, (A.4) and (A.5) together are conditions for two positive eigenvalues, and lastly, given the other three conditions, (A.6) is the condition for two stable eigenvalues.

Condition (A.3) holds if either $\alpha_1 > (\sqrt{1-\delta} + \sqrt{\alpha_2})^2$ or $\alpha_1 < (\sqrt{1-\delta} - \sqrt{\alpha_2})^2$. Note that the latter is precisely Assumption A1(ii). Next, condition (A.4) is equivalent to $\alpha_1 < 1 - \delta + \alpha_2$. This is violated under the first possible condition ensuring (A.3), but is implied by the second. Thus, Assumption A1(ii) is necessary and sufficient to have both (A.3) and (A.4) hold. Next, from (A.2) and the basic parameter restrictions, condition (A.5) clearly holds. Finally, condition (A.6) is equivalent to Assumption A1(i). Thus, Assumption A1 is necessary and sufficient to ensure that both eigenvalues are real, positive, and stable. □

Proposition 2

The period-t symmetric equilibrium is given by the level of effort e_t that solves

$$e_t = \max\left\{\alpha_t + F(e_t), 0\right\}, \tag{A.7}$$

where $\alpha_t = \alpha_0 - \alpha_1 X_t + \alpha_2 e_{t-1}$ is a given history. The left-hand side of equation (A.7) is strictly increasing, with slope equal to one. Suppose $\alpha_t + F(0) \leq 0$. Then clearly $e_t = 0$ is an equilibrium. Furthermore, since $F'(e) \leq \rho < 1$ by Assumptions A2-A3, the expression $\alpha_t + F(e_t)$ remains below the 45-degree line for $e_t > 0$. Thus, $e_t = 0$ is the unique equilibrium for this case.

Suppose instead $\alpha_t + F(0) > 0$. In this case, for e_t sufficiently small $\alpha_t + F(e_t)$ will be above the 45-degree line. Further, since $F'(e) \leq \rho < 1$, $\alpha_t + F(e_t)$ must eventually cross the 45-degree line at some $e_t = e_t^* > 0$, and then remain below the 45-degree line thereafter. Thus, $e_t = e_t^*$ is the unique equilibrium for this case. □

Proposition 3

First, recall that Assumption A1 implies $\rho^* > 0$. Further, clearly $\rho^* < 1$ if and only if $\alpha_1 < \delta\alpha_2$. Next, combining the two equations in (5), a SS value of e is any solution to

$$\rho^* e = \alpha_0 + F(e) \equiv G(e)\,. \tag{A.8}$$

If $\rho = 0$, this condition becomes

$$\rho^* e = \alpha_0\,.$$

which is uniquely satisfied by $e = e^s$.

Thus, suppose $\rho \neq 0$. The left-hand side of (A.8) is a straight line emanating from the origin with slope ρ^*. That slope is positive and smaller than one if $\alpha_1 < \delta\alpha_2$, and larger than one otherwise. Meanwhile, the shape of $G(e)$ (the right-hand side of (A.8)) is determined by the properties of F. Note that the vertical intercept $G(0) = \alpha_0 + F(0)$ is strictly positive by Assumption A2(iii), so that G always begins above the $\rho^* e$ line, and by Assumptions A2(iv), A2(vi), and A3, we have $G'(e) = F'(e) \leq G'(e^s) = \rho < 1$, where the weak inequality is strict for $e \neq e^s$.

If $\rho < 0$ (see panel (a) in Figure A.1), then G is strictly decreasing, and will therefore intersect the $\rho^* e$ line exactly once, so that the SS is unique. Similarly, if $0 < \rho \leq \rho^*$, then G is increasing but flatter than the $\rho^* e$ line (except possibly exactly at $e = e^s$), so that again the two intersect exactly once, and therefore there is a unique SS. If $\alpha_1 \geq \delta\alpha_2$, then $\rho^* \geq 1 > \rho \geq G'(e)$ (Figure A.1(b)), so that we are necessarily in this case, which completes the proof of part (i) of the Proposition.

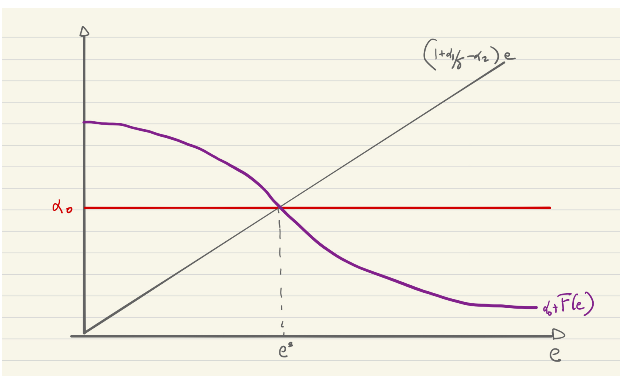
If $\alpha_1 < \delta\alpha_2$, meanwhile, we have $\rho^* < 1$, so that it is possible to have $G'(e) > \rho^*$. As already established, if $\rho \leq \rho^*$ (Figure A.1(c)) then this does not happen. If, however, $\rho > \rho^*$ (Figure A.1(d)), then since $G(e^s) = e^s$ and $G'(e^s) = \rho$, it follows that G intersects the $\rho^* e$ line at e^s from below. Since, as already noted, G is necessarily above the $\rho^* e$ line for e sufficiently small, this implies the existence of another intersection—and therefore another SS—at a value $e < e^s$. Meanwhile, by Assumption A2(v), G must eventually become flatter than the $\rho^* e$ line, implying the existence of a third intersection/SS at a value $e > e^s$. Finally, by Assumption A2(vi), the slope of G monotonically decreases as e moves away from e^s in either direction, and therefore there can only be one additional intersection/SS below e^s and one above e^s, making exactly three SSs total. □

Proposition 4

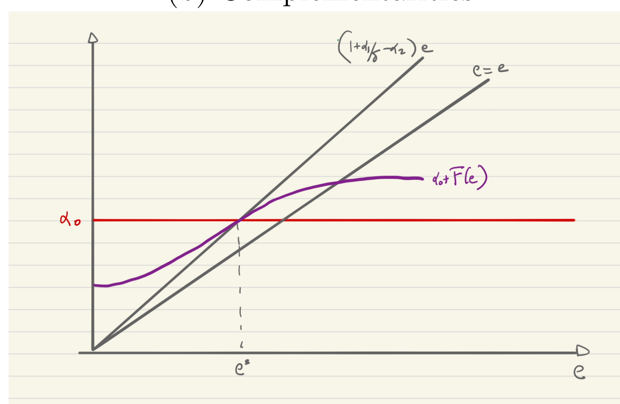
As stated in the text, as ρ decreases from 0, (T, D) moves continuously along the line (11) starting from the no-interaction point E_0 (see Figure 4) and approaching $E_{-\infty} \equiv (1-\delta, 0)$ as $\rho \to -\infty$. Since E and $E_{-\infty}$ both lie in the intersection of triangle $\widehat{ABC}$ (which is the region featuring two stable eigenvalues) with the upper-right quadrant of the (T, D) plane (which

Figure A.1: Steady state(s)

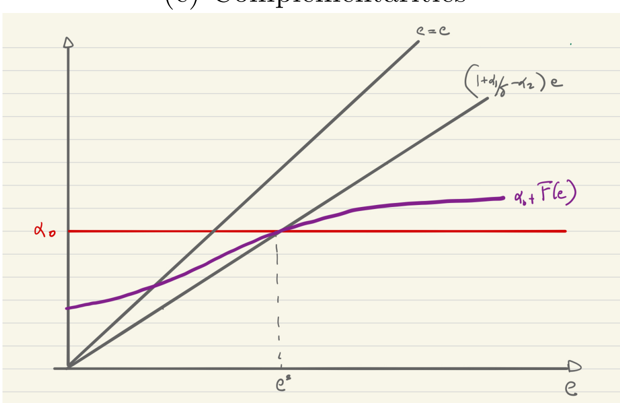
(a) Substitutabilities



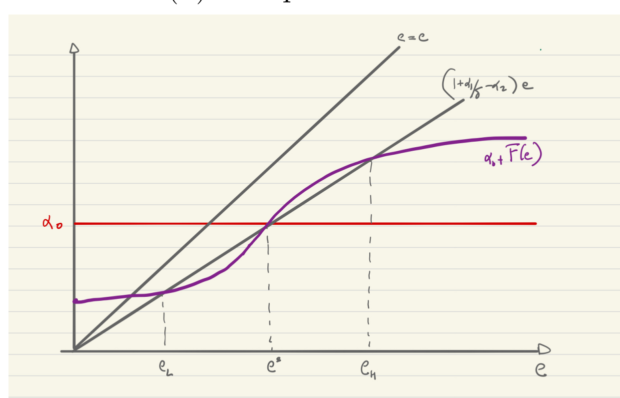
(b) Complementarities



(c) Complementarities



(d) Complementarities



Notes: This Figure depicts the solutions to the equation $\left(1+\frac{\alpha_1}{\delta}-\alpha_2\right)e=\alpha_0+F(e)$ *that is satisfied at a SS of the model.*

is the region where both eigenvalues have positive real parts), and since that intersection is convex, it follows that (T, D) must remain in this "both stable with positive real parts" region for all values of $\rho \leq 0$. This confirms parts (i) and (ii) of the Proposition.

Next, in addition to the above properties, E_0 and $E_{-\infty}$ both also lie below the parabola $D = T^2/4$, and therefore for $\rho \leq 0$ sufficiently large or sufficiently small, the eigenvalues must also be real and positive. If $\alpha_1 \geq \alpha_2$, then line (11) is vertical or negatively sloped, in which case the entire line segment connecting E and $E_{-\infty}$ must also lie below the parabola, and therefore the eigenvalues are always real and positive for $\rho \leq 0$.

Suppose instead that $\alpha_1 < \alpha_2$. Then there is a part of line (11) lying above the $D = T^2/4$ parabola if and only if there are values of T such that $\psi(T) < 0$, where

$$\psi(T) \equiv \frac{1}{4}T^2 - \frac{\alpha_2 (1-\delta)}{\alpha_2 - \alpha_1}T + \frac{\alpha_2 (1-\delta)^2}{\alpha_2 - \alpha_1}.$$

$\psi(T)$ is a convex parabola with roots

$$T_1 \equiv 2\frac{(1-\delta)\sqrt{\alpha_2}}{\sqrt{\alpha_2}+\sqrt{\alpha_1}} \quad \text{and} \quad T_2 \equiv 2\frac{(1-\delta)\sqrt{\alpha_2}}{\sqrt{\alpha_2}-\sqrt{\alpha_1}},$$

so that $\psi(T) < 0$ if and only if T_1, T_2 are real and $T_1 < T < T_2$. If $\alpha_1 < 0$ then the roots are complex, while if $\alpha_1 = 0$ then $T_1 = T_2$ and therefore we cannot have $T_1 < T < T_2$. Thus, if $\alpha_1 \leq 0$, then no part of the line (11) can lie above the parabola, and therefore (T, D) must remain real (and positive) as for all $\rho \leq 0$.

Suppose then that $0 < \alpha_1 < \alpha_2$, so that T_1, T_2 are real with $T_1 < T_2$. Using (9), the condition $T_1 < T < T_2$ can be equivalently expressed as $\rho \in (\rho_1, \rho_2)$, where

$$\rho_1 \equiv 1 - \frac{\left(\sqrt{\alpha_2}+\sqrt{\alpha_1}\right)^2}{1-\delta}, \qquad \rho_2 \equiv 1 - \frac{\left(\sqrt{\alpha_2}-\sqrt{\alpha_1}\right)^2}{1-\delta}.$$

Suppose $\alpha_2 \leq 1-\delta$. Then by Assumption A1(ii), $(\sqrt{\alpha_1}+\sqrt{\alpha_2})^2 < 1-\delta$, in which case $\rho_1 > 0$. Thus, the part of line (11) lying above the parabola corresponds to values of $\rho > 0$, and thus for $\rho \leq 0$ the line lies below the parabola, so that again (T, D) remains real for all $\rho \leq 0$. This completes the proof of part (iii).

Finally, suppose again that $0 < \alpha_1 < \alpha_2$, but now $\alpha_2 > 1-\delta$. Then by Assumption A1(ii), $(\sqrt{\alpha_1}-\sqrt{\alpha_2})^2 > 1-\delta$, in which case $\rho_2 < 0$. Thus, we have $-\infty < \rho_1 < \rho_2 < 0$, and the part of line (11) corresponding to $\rho \in (\rho_1, \rho_2)$ lies above the parabola, so that the eigenvalues are complex for this range of negative values of ρ. This completes the proof of part (iv). □

Proposition 5

As noted in the text, as ρ increases from 0 to 1, (T, D) moves upward along the line (11) from point E_0, eventually exiting triangle $\widehat{ABC}$ at some point $\rho = \bar{\rho}$. We have a flip bifurcation (eigenvalue equal to -1) if this exit occurs via the left edge of the triangle, a Hopf bifurcation (pair of complex eigenvalues with modulus 1) if it occurs via the top edge, and a fold bifurcation (eigenvalue equal to 1) if it occurs via the right edge.

A flip bifurcation thus occurs if and only if line (11) is negatively sloped, with $D < 1$ at the point on the line where $T = -2$ (see Figure 5). The first condition is met if and only if $\alpha_2 < \alpha_1$. Assuming this is the case, the second condition occurs if and only if

$$\frac{\alpha_2(1-\delta)}{\alpha_2 - \alpha_1}\left[-2-(1-\delta)\right] < 1\,,$$

which can be simplified to the condition given in part (i) of the Proposition. Further, since $2 - \delta > 1$, this latter condition on its own actually implies the condition $\alpha_2 < \alpha_1$ for a negative slope, so that the condition given in part (i) is, on its own, necessary and sufficient for a flip bifurcation.

Next, a fold bifurcation occurs if and only if line (11) is positively sloped, with $D < 1$ at the point on the line where $T = 2$. The first condition is met if and only if $\alpha_2 > \alpha_1$. Assuming this is the case, the second condition occurs if and only if

$$\frac{\alpha_2(1-\delta)}{\alpha_2 - \alpha_1}\left[2-(1-\delta)\right] < 1\,,$$

which can be simplified to the condition from part (iii) of the Proposition. Since $\delta < 1$, that condition on its own implies the condition $\alpha_2 > \alpha_1$ for a positive slope, and therefore again the condition given in part (iii) is, on its own, necessary and sufficient for a fold bifurcation.

Finally, a Hopf bifurcation occurs if and only if either (a) line (11) is negatively sloped, with $D > 1$ at the point on the line where $T = -2$; (b) line (11) is positively sloped, with $D > 1$ at the point on the line where $T = 2$; or (c) line (11) is vertical. Given the analysis for the flip and fold cases, (a) occurs if and only if $\frac{\alpha_1}{(2-\delta)^2} < \alpha_2 < \alpha_1$, (b) occurs if and only if $\alpha_1 < \alpha_2 < \frac{\alpha_1}{\delta^2}$, while (c) occurs if and only if $\alpha_1 = \alpha_2$. It is straightforward to see that one of these cases must occur if and only if the condition given in part (ii) of the Proposition holds. □

Proposition 6

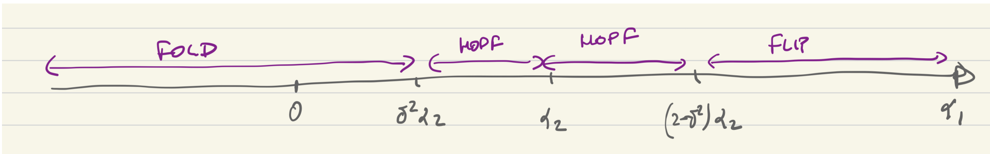
First, note that the 2-cycle referenced in the flip case and the closed invariant curve referenced in the Hopf case are well established general properties of such bifurcations (see, e.g., Guckenheimer and Holmes [2002]) whenever the system is locally non-linear (as Assumption A2(vi) ensures in this case). It thus remains only to verify the claims about SSs.

Based on the conditions established in Proposition 5, Figure A.2 shows, for given α_2 and δ, the values of α_1 corresponding to flip, Hopf, and fold bifurcations.

A necessary condition for the existence of multiple SSs is $\alpha_1 < \delta\alpha_2$ (see Proposition 3), a necessary condition for which is $\alpha_1 < \alpha_2$. Let us therefore restrict to the $(-\infty, \alpha_2)$ half-line in Figure A.2. In this case, only Hopf and fold bifurcations are possible, which confirms that the SS remains unique for the flip case.

Next, the Hopf bifurcation portion of this half-line corresponds to $\delta^2\alpha_2 < \alpha_1 < \alpha_2$. For the $\delta\alpha_2 \le \alpha_1 < \alpha_2$ part of this, Proposition 3 rules out additional SSs. Thus, consider the $\delta^2\alpha_2 < \alpha_1 < \delta\alpha_2$ portion. Recall from Proposition 3 that $\rho^* = 1 - \alpha_2 + \alpha_1/\delta$ is the value of ρ above which we will have multiple SSs. Meanwhile, for a Hopf bifurcation the bifurcation value $\bar{\rho}$ is such that $D = 1$, and therefore from (10) we have $\bar{\rho} = 1 - \alpha_2(1-\delta)$. Thus, there

Figure A.2: Bifurcations on the α_1 line



Notes: For given δ and α_2, we show on the α_1 line the conditions for fold, Hopf and flip bifurcations.

will not be multiple SSs at the Hopf bifurcation if and only if $\bar{\rho} < \rho^*$, which is equivalent to the condition $\alpha_1 > \alpha_2\delta^2$, a condition that we have already taken to hold by hypothesis. Thus, we conclude that the SS must remain unique for the flip case as well.

Finally, for the fold case we must have $\alpha_1 < \delta^2\alpha_2$. The bifurcation in this case occurs at the value $\bar{\rho}$ of ρ at which line (11) (which is positively sloped in this case) intersects the $\Delta = T - 1$ line. At this intersection, we have

$$T = \frac{\alpha_2\delta\,(2-\delta) - \alpha_1}{\alpha_2\delta - \alpha_1}\,,$$

and therefore from (9) one can solve to obtain $\bar{\rho} = \rho^*$. Thus, from Proposition 3, the bifurcation point is precisely the point at which two additional SSs must appear. $\square$

Proposition 7

For symmetric allocations, and letting hats denote deviations from the no-interaction SS, our non-linear dynamical system can be written

$$H\left(\hat{e}_t\right) = \left(\alpha_2 - \alpha_1\right)\hat{e}_{t-1} - \alpha_1\left(1-\delta\right)\widehat{X}_{t-1}\,, \tag{A.9}$$

$$\widehat{X}_t = \left(1-\delta\right)\widehat{X}_{t-1} + e_{t-1}\,, \tag{A.10}$$

where $H(\hat{e}) = \hat{e} - \widehat{F}(\hat{e})$ and $\widehat{F}(\hat{e}) \equiv F(e^s + \hat{e})$. Note that since $F(e^s) = 0$ by Assumption A2(ii), we have $H(0) = \widehat{F}(\hat{e}) = 0$.

Hopf bifurcation case

Since $\widehat{F}'(\hat{e}) = F'(e^s + \hat{e}) \leq \rho$ by Assumption A2(iv), it follows that, for $\rho < 1$ (which is the relevant case here), H is a strictly increasing function, and is therefore invertible. Letting $G(\cdot) \equiv H^{-1}(\cdot)$ denote its inverse, we can write (A.9) as

$$\hat{e}_t = G\left(\left(\alpha_2 - \alpha_1\right)\hat{e}_{t-1} - \alpha_1\left(1-\delta\right)\widehat{X}_{t-1}\right)\,.$$

Next, note that $G(0) = 0$ and $G'(0) = 1/(1-\rho)$. Thus, we can write this as

$$\hat{e}_t = \frac{\alpha_2 - \alpha_1}{1-\rho}\hat{e}_{t-1} - \frac{\alpha_1\left(1-\delta\right)}{1-\rho}\widehat{X}_{t-1} + m\left(\hat{e}_{t-1}, \widehat{X}_{t-1}\right)\,,$$

where

$$m\left(\hat{e},\widehat{X}\right)\equiv G\left(\left(\alpha_2-\alpha_1\right)\hat{e}-\alpha_1\left(1-\delta\right)\widehat{X}\right)-\frac{\alpha_2-\alpha_1}{1-\rho}\hat{e}+\frac{\alpha_1\left(1-\delta\right)}{1-\rho}\widehat{X}\,,$$

and by construction m is $O(\|(\hat{e},\widehat{X})\|^2)$ as $(\hat{e},\widehat{X})\to 0$ (i.e., the Taylor expansion of m around 0 contains no constant or linear terms). Thus, the bivariate system (A.9)-(A.10) can be written as

$$\begin{pmatrix}\hat{e}_t\\ \widehat{X}_t\end{pmatrix}=M\begin{pmatrix}\hat{e}_{t-1}\\ \widehat{X}_{t-1}\end{pmatrix}+\begin{pmatrix}m\left(\hat{e}_{t-1},\widehat{X}_{t-1}\right)\\ 0\end{pmatrix}\,, \tag{A.11}$$

where M is the first-order matrix defined in (6). Note that M contains all first-order information for this system (with all higher-order information contained in m).

To prove the desired result, we make use of Wan's [1978] theorem, and of the formulation of it given by Wikan [2013]. In what follows, we take ρ to be in a sufficiently small neighborhood of $\bar{\rho}$ so that the eigenvalues of the linearized system are complex conjugates. For a given ρ, let $\lambda(\rho)$ denote the eigenvalue with positive imaginary part, and $\lambda\equiv\lambda(\bar{\rho})$ (i.e., the eigenvalue at the bifurcation point).

To study the stability of the limit cycle as the system goes through a Hopf bifurcation, we first need to convert system (A.11) at the bifurcation point (i.e., for $\rho=\bar{\rho}$) into the "standard form"

$$\begin{pmatrix}y_{1t}\\ y_{2t}\end{pmatrix}=\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}y_{1t-1}\\ y_{2t-1}\end{pmatrix}+\begin{pmatrix}f(y_{1t-1},y_{2t-1})\\ g(y_{1t-1},y_{2t-1})\end{pmatrix}\,, \tag{A.12}$$

where $\theta\in(0,\pi)$, $(y_1,y_2)=\zeta(\hat{e},\widehat{X})$ for some smooth invertible function ζ whose inverse ζ^{-1} is also smooth and for which $\zeta(0)=0$, and f and g are $O(\|(y_1,y_2)\|^2)$ as $(y_1,y_2)\to 0$. Given the model written in this standard form, define

$$a=-\mathrm{Re}\left(\frac{(1-2\lambda)\overline{\lambda}^2}{1-\lambda}\xi_{11}\xi_{20}\right)-\frac{1}{2}|\xi_{11}|^2-|\xi_{02}|^2+\mathrm{Re}\left(\overline{\lambda}\xi_{21}\right)\,,$$

where $\overline{x}$ and $\mathrm{Re}(x)$ indicate the complex conjugate and real part of x, respectively, and

$$\begin{aligned}
\xi_{20}&=\frac{1}{8}\left[(f_{11}-f_{22}+2g_{12})+i\,(g_{11}-g_{22}-2f_{12})\right]\,,\\
\xi_{11}&=\frac{1}{4}\left[(f_{11}+f_{22})+i\,(g_{11}+g_{22})\right]\,,\\
\xi_{02}&=\frac{1}{8}\left[(f_{11}-f_{22}-2g_{12})+i\,(g_{11}-g_{22}+2f_{12})\right]\,,\\
\xi_{21}&=\frac{1}{16}\left[(f_{111}+f_{122}+g_{112}+g_{222})+i\,(g_{111}+g_{122}-f_{112}-f_{222})\right]\,.
\end{aligned}$$

As shown in Wikan [2013] (see Theorem 2.5.2), the Hopf bifurcation is supercritical if $\partial|\lambda(\rho)|/\partial\rho\big|_{\rho=\bar{\rho}}>0$ and $a<0$. We have already verified the first of these conditions (see Proposition 5), so it remains only to check the second.

By definition, the eigenvalues of M at the Hopf bifurcation point are complex and satisfy $|\lambda| = 1$, so that we may write $\lambda = \cos\theta + i\sin\theta$, $\overline{\lambda} = \cos\theta - i\sin\theta$, for some $\theta \in (0, \pi)$. Define

$$C = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} .$$

By construction, λ and $\overline{\lambda}$ are the eigenvalues of C. Define also

$$R = \begin{pmatrix} \lambda & 0 \\ 0 & \overline{\lambda} \end{pmatrix} , \qquad V_C = \begin{pmatrix} \sin\theta & \sin\theta \\ -i\sin\theta & i\sin\theta \end{pmatrix} , \qquad V_M = \begin{pmatrix} \lambda + \delta - 1 & \overline{\lambda} + \delta - 1 \\ 1 & 1 \end{pmatrix} .$$

It can be verified that the columns of V_C and V_M are eigenvectors of C and M, respectively, so that $C = V_C R V_C^{-1}$ and $M = V_M R V_M^{-1}$. We therefore have $M = V_M V_C^{-1} C V_C V_M^{-1} = B^{-1} C B$, where

$$B \equiv V_C V_M^{-1} = \begin{pmatrix} 0 & \sin\theta \\ -1 & \cos\theta - (1-\delta) \end{pmatrix} .$$

Thus, making the change of variables $(y_1, y_2)' = B \times (\hat{e}, \widehat{X})'$, we may write (A.11) for the $\rho = \bar{\rho}$ case in the standard form (A.12), where

$$\begin{aligned} f(y_1, y_2) &= 0 , \\ g(y_1, y_2) &= \frac{\gamma_1}{\sin\theta} y_1 - \gamma_2 y_2 - G\left(\frac{\gamma_1}{\sin\theta} y_1 - \gamma_2 y_2\right) , \end{aligned}$$

and $\gamma_1 \equiv (\alpha_2 - \alpha_1)\cos\theta - \alpha_2(1-\delta)$, $\gamma_2 \equiv \alpha_2 - \alpha_1$.

We can now check the condition $a < 0$ for the Hopf bifurcation to be supercritical. Since $\bar{\rho} \in (0, 1)$, Assumptions A2(i) and A2(vi) imply that $F'' = 0$ and $F''' < 0$ (where derivatives without arguments indicate values at the no-interaction SS), and therefore

$$G'' = 0 , \qquad G''' = \frac{F'''}{(1-\bar{\rho})^4} < 0 .$$

Thus, $g_{11} = g_{12} = g_{22} = 0$, and since $f(y_1, y_2) = 0$, it follows that $\xi_{20} = \xi_{11} = \xi_{02} = 0$ and thus

$$a = \mathrm{Re}\left(\overline{\lambda}\xi_{21}\right) = \frac{\alpha_2 (1-\delta)}{16} \left(\frac{\gamma_1^2}{\sin^2\theta} + \gamma_2^2\right) G''' < 0 .$$

This confirms that the limit cycle is supercritical. □

Flip bifurcation case

A flip bifurcation occurs at the intersection of line (11) and the line $D = -T - 1$. It can be verified that this intersection occurs for

$$\rho = \bar{\rho} = 1 + \alpha_2 - \frac{\alpha_1}{2-\delta} \in (0, 1) . \tag{A.13}$$

In what follows, we consider only values of ρ in a neighborhood of $\bar{\rho}$.

Let λ be the smallest eigenvalue of M. Since $\rho \approx \bar{\rho}$, $\lambda \approx -1$. An eigenvector of M associated with this λ is $v = (\lambda + \delta - 1, 1)'$, so that the associated eigenspace can be written $\zeta = \{\beta v : \beta \in \mathbb{R}\}$. Tangent to this one-dimensional eigenspace at $(\hat{e}, \widehat{X}) = (0, 0)$ is a (locally) analytic one-dimensional invariant manifold μ. For some open interval $\Xi \subset \mathbb{R}$ containing 0, we can write $\mu = \{(\phi(\widehat{X}), \widehat{X}) : \widehat{X} \in \Xi\}$ for an analytic function ϕ, where tangency to ζ implies $\phi' = \lambda + \delta - 1$ (where derivatives without arguments are evaluated at $\widehat{X} = 0$).

Since μ is invariant, and $(\hat{e}, \widehat{X}) \in \mu$ implies $\hat{e} = \phi(\widehat{X})$, (A.9)-(A.10) together imply that ϕ implicitly solves

$$H\left(\phi\left((1-\delta)\widehat{X} + \phi\left(\widehat{X}\right)\right)\right) = (\alpha_2 - \alpha_1)\,\phi\left(\widehat{X}\right) - \alpha_1 (1-\delta)\widehat{X}\,. \tag{A.14}$$

Differentiating (A.14) twice and evaluating at the SS, we obtain

$$\left(\lambda^2 + \lambda - T\right)\phi'' = 0\,,$$

where T is as usual the trace of M, and we have used the facts that $\phi' = \lambda + \delta - 1$, $H' = 1 - F' = 1 - \rho$, and $H'' = -F'' = 0$ (where the last equality is implied by Assumptions A2(i) and A2(vi)). Since, for ρ near $\bar{\rho}$ we have $T < 0$ and $\lambda \approx -1$, the term in parentheses is non-zero, and therefore $\phi'' = 0$.

Next, differentiating (A.14) three times and evaluating at the SS, we may solve for

$$\phi''' = \frac{(\lambda + \delta - 1)^3\,\lambda^3 F'''}{(1-\rho)\left(\lambda^3 + \lambda + \delta - 1\right) - (\alpha_2 - \alpha_1)}\,, \tag{A.15}$$

where we have used the values of ϕ' and ϕ'' obtained above, along with the facts that $H' = 1 - \rho$, $H'' = 0$, and $H''' = -F'''$.

The evolution of the restriction of (A.9)-(A.10) to μ can be written as $\widehat{X}_t = \chi(\widehat{X}_{t-1}) \equiv (1-\delta)\widehat{X}_{t-1} + \phi(\widehat{X}_{t-1})$, with $\hat{e}_t = \phi(\widehat{X}_t)$. Note that, for $\rho = \bar{\rho}$, we have $\chi' = -1$. From Wikan [2013] (see Theorem 1.5.1), the flip bifurcation in the system $\widehat{X}_t = \chi(\widehat{X}_{t-1})$ at $\rho = \bar{\rho}$ is supercritical if, for the case where $\rho = \bar{\rho}$, we have $b > 0$, where

$$b \equiv \frac{1}{2}\chi''^2 + \frac{1}{3}\chi'''\,.$$

Since $\chi'' = \phi'' = 0$, we need only verify that $\chi''' = \phi''' > 0$ when $\rho = \bar{\rho}$. Substituting the value of $\bar{\rho}$ from (A.13) into (A.15) for ρ, and using the fact that $\lambda = -1$ for this case, we may obtain that

$$\phi''' = -\frac{(2+\delta)^3\,(2-\delta)\,F'''}{\alpha_1 - \alpha_2\,(2-\delta)^2}\,.$$

Proposition 5 implies that the denominator of this expression is positive, while since $\rho = \bar{\rho} > 0$, Assumption A2(vi) implies that $F''' < 0$. We therefore conclude that $\chi''' = \phi''' > 0$, and thus the flip bifurcation is supercritical. □

Proposition 8

Linearizing the system around the SS (e_j, X_j), $j = L, H$, the first-order matrix is identical to M from (6), except with $F'(e_j)$ in place of ρ. Thus, analysis of the stability of the SS (e_j, X_j) is identical to the analysis of the stability of (e^s, X^s), except with $F'(e_j)$ in place of ρ. Using directly analogous arguments to the fold case from the proof of Proposition 6, we conclude that both eigenvalues are stable as long as $F'(e_j) < \bar{\rho} = \rho^* \equiv 1 - \alpha_2 + \alpha_1/\delta$. From panel (d) of Figure A.1, meanwhile, we observe that indeed this must be the case, and therefore (e_j, X_j) is stable for $j = L, H$. □

A.2 The Forward-Looking Model

Preliminaries

We begin by defining some useful notation. Letting $\phi \equiv 1-\rho$, the characteristic polynomial of M for $e^* = e^s$ is given by

$$P(\lambda) \equiv \lambda^3 - \left(1-\delta+\frac{\phi}{\alpha_3}\right)\lambda^2 + \frac{\phi(1-\delta)+\alpha_2-\alpha_1}{\alpha_3}\lambda - (1-\delta)\frac{\alpha_2}{\alpha_3}\,, \tag{A.16}$$

so that the eigenvalues of M are the roots of M. It will be useful in some cases to write P as

$$P(\lambda) = (\lambda-\lambda_2)\left(\lambda^2-\tau\lambda+c\right)\,, \tag{A.17}$$

where the roots of the quadratic polynomial $\lambda^2-\tau\lambda+c$ are λ_{11} and λ_{12}, and $\lambda_{11}, \lambda_{12}, \lambda_2$ are as defined in the main text. In particular, note that

$$\lambda_{11} = \frac{\tau-\sqrt{\tau^2-4c}}{2} \quad , \quad \lambda_{12} = \frac{\tau+\sqrt{\tau^2-4c}}{2}\,, \tag{A.18}$$

and λ_2, τ, and c satisfy the relationships

$$\tau+\lambda_2 = 1-\delta+\frac{\phi}{\alpha_3}\,, \tag{A.19}$$

$$c+\tau\lambda_2 = \frac{\phi(1-\delta)+\alpha_2-\alpha_1}{\alpha_3}\,, \tag{A.20}$$

$$c\lambda_2 = (1-\delta)\frac{\alpha_2}{\alpha_3}\,. \tag{A.21}$$

Also, $\tau = \lambda_{11}+\lambda_{12}$ and $c = \lambda_{11}\lambda_{12}$.

Note further that

$$P(1) = -\frac{\alpha_1+(\phi-\alpha_2-\alpha_3)\delta}{\alpha_3}\,, \tag{A.22}$$

and that

$$P(0) = -\frac{(1-\delta)\alpha_2}{\alpha_3} < 0\,, \tag{A.23}$$

$$P'(0) = \frac{\phi(1-\delta)+\alpha_2-\alpha_1}{\alpha_3}\,, \tag{A.24}$$

$$P''(0) = -\frac{2\left[(1-\delta)\alpha_3+\phi\right]}{\alpha_3} < 0\,. \tag{A.25}$$

Since $P(0)<0$ and $P''(0)<0$, if P has three real roots and $P'(0)>0$, then all three roots must be positive. If P has three real roots but instead $P'(0)<0$, then two of the roots must be negative and the third positive.

Proposition 9

When $\rho = 0$, we have $\phi = 1$. Under Assumption A4(i), $P(1) < 0$ for this case (see (A.22)). Also, since $P''' > 0$, we must have $P(\lambda) > 0$ for λ large enough. Thus, there must be a $\lambda \in (1, \infty)$ with $P(\lambda) = 0$, and therefore at least one of the real roots of P is greater than 1. Since λ_2 is, by definition, the largest real root of P, this implies $\lambda_2 > 1$.

Next, note that, for the $\phi = 1$ case, we can equivalently write the linearized system (14) as $Ux_{t+1} = Vx_t$, where $x_t \equiv (X_t, e_t, e_{t-1})'$,

$$U \equiv \begin{pmatrix} 1 & 0 & 0 \\ 0 & \alpha_3 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \text{and} \qquad V \equiv \begin{pmatrix} 1-\delta & 1 & 0 \\ \alpha_1 & \phi & -\alpha_2 \\ 0 & 1 & 0 \end{pmatrix}.$$

The generalized eigenvalues of the matrix pencil $U\lambda - V$ (i.e., the values of λ such that $|U\lambda - V| = 0$) are given by the roots of

$$R(\lambda; \alpha_3) \equiv \alpha_3 \lambda^3 - [\alpha_3 (1-\delta) + 1] \lambda^2 + [(1-\delta) + \alpha_2 - \alpha_1] \lambda - (1-\delta) \alpha_2 .$$

As is well known (see, e.g., Klein [2000]), there is a tight relationship between these generalized eigenvalues and the eigenvalues of M (i.e., the roots of $P(\lambda)$). In particular, if $\alpha_3 > 0$, so that U is non-singular, then the three generalized eigenvalues of $U\lambda - V$ and the eigenvalues of M are the same. In the case where $\alpha_3 = 0$, meanwhile, there are only two generalized eigenvalues of $U\lambda - V$. In this case, we refer to the "missing" eigenvalue as infinite, since $\lim_{\alpha_3 \to 0} \lambda_2 = \infty$.[17]

The remaining two generalized eigenvalues, $\lambda_{11}, \lambda_{12}$, do not disappear at $\alpha_3 = 0$, and therefore they are continuous functions of α_3 at that point. We therefore conclude that, as $\alpha_3 \to 0$, the roots $\lambda_{11}, \lambda_{12}$ of P converge to the two roots of $R(\lambda; 0)$. It is straightforward to verify that $R(\lambda; 0) = -Q(\lambda)$, where Q is the characteristic polynomial from the backward-looking model (see (7)), and therefore $R(\lambda; 0)$ and $Q(\lambda)$ have the same roots. Thus, as $\alpha_3 \to 0$, $\lambda_{11}, \lambda_{12}$ converge to the eigenvalues for the backward-looking model. Since Assumption A4 coincides with A1 for this limiting case, by Proposition 1 we conclude that, for α_3 close to zero, the eigenvalues for the no-interaction case are both real, positive and stable.

Next, note that $R(0; \alpha_3) = -(1-\delta)\alpha_2 < 0$, $R_\lambda(0; \alpha_3) = 1 - \delta + \alpha_2 - \alpha_1 > 0$ (by Assumption A4(ii)), and $R_{\lambda\lambda}(0; \alpha_3) = -2[\alpha_3(1-\delta) + 1] < 0$. Thus, at $\lambda = 0$, the cubic function $R(\cdot; \alpha_3)$ is negative, upward-sloping, and concave. Since $R_{\lambda\lambda\lambda}(\lambda; \alpha_3) = 6\alpha_3 > 0$, it follows that any real roots of R must be greater than zero.

Next, $R(1; \alpha_3) = -[\alpha_1 + (1 - \alpha_2 - \alpha_3)\delta] < 0$ (by Assumption A4(i)). Thus, since $R(0; \alpha_3) < 0$ as well, if $R(\lambda; \alpha_3) \geq 0$ for some $\lambda \in (0, 1)$, we must have $\lambda_{11}, \lambda_{12} \in (0, 1)$. Now, $R(1-\delta; \alpha_3) = -\alpha_1(1-\delta)$. If $\alpha_1 \leq 0$, then indeed $R(1-\delta; \alpha_3) \geq 0$, and therefore $\lambda_{11}, \lambda_{12} \in (0, 1)$ for all $\alpha_3 \leq 1$. Thus, for the $\alpha_1 \leq 0$ case we may set $\bar{\alpha}_3 = 1$ in the statement of the Proposition.

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[17]One can check this by noting that, for λ large enough, $R(\lambda; 0) < 0$. Let $\widetilde{\lambda}$ be such a value. Then there then exists a sufficiently small $\alpha_3 > 0$ such that $R(\widetilde{\lambda}; \alpha_3) < 0$. Since $R''' = \alpha_3 > 0$ in this case, however, $R(\lambda, \alpha_3) > 0$ for λ sufficiently large, and therefore $R(\lambda, \alpha_3) = 0$ for some $\lambda > \widetilde{\lambda}$. Since $\widetilde{\lambda}$ was chosen arbitrarily, it follows that we can always find a real generalized eigenvalue as large as we want by choosing $\alpha_3 > 0$ sufficiently small, and therefore $\lambda_2 \to \infty$ as $\alpha_3 \to 0$.

Suppose instead that $\alpha_1 > 0$, so that $R(1-\delta;\alpha_3) < 0$. In this case,

$$R_\lambda(1-\delta;\alpha_3) = -\left\{\left[1-\alpha_3\left(1-\delta\right)\right]\left(1-\delta\right)+\alpha_1\right\} < -\left[\delta\left(1-\delta\right)+\alpha_1\right] < 0\,,$$

where the second inequality follows from the parameter restriction $\alpha_3 < 1$. Thus, when $\alpha_1 > 0$, at $\lambda = 1-\delta$ the cubic function $R(\cdot;\alpha_3)$ is downward-sloping and negative. Since $R_{\lambda\lambda\lambda} > 0$, this implies that if $\lambda_{11}, \lambda_{12}$ are real then they must be less than $1-\delta$. Combined with our earlier result that any such real roots must be greater than zero, we conclude that if $\alpha_1 > 0$, then if $\lambda_{11}, \lambda_{12}$ are real they must be in $(0, 1-\delta)$. For α_3 sufficiently close to zero, we already established that these roots are indeed real, and thus $\max_{\lambda\in(0,1-\delta)} R(\lambda;\alpha_3) > 0$ for $\alpha_3 > 0$ sufficiently small. Note that $R_{\alpha_3}(\lambda;\alpha_3) = \lambda^3 - (1-\delta)\lambda^2$, which is strictly negative for $\lambda \in (0, 1-\delta)$. Thus, the segment of $R(\lambda;\alpha_3)$ corresponding to $\lambda \in (0, 1-\delta)$ shifts downward as α_3 increases. For the $\alpha_1 > 0$ case, set $\bar{\alpha}_3$ in the statement of the Proposition to be the maximum value of $\alpha_3 \in (0,1]$ such that $\max_{\lambda\in(0,1-\delta)} R(\lambda;\bar{\alpha}_3) \geq 0$. Then for $\alpha_3 \leq \bar{\alpha}_3$, we necessarily have $\lambda_{11}, \lambda_{12} \in (0, 1-\delta)$, while for $\alpha_3 \in (\bar{\alpha}_3, 1)$ there are no real roots in $(0, 1-\delta)$, and therefore $\lambda_{11}, \lambda_{12}$ cannot be real. This completes the proof. □

Proposition 10

The proof is trivially adapted from the proofs of Propositions 2 and 3 (which correspond to the case $\alpha_3 = 0$). □

Proposition 11

Note that we are interested only in the case where $\rho \leq 0$, which is equivalent to $\phi \geq 1$. Assume henceforth that this is the case. Under Assumption A4(i), this implies $P(1) < 0$ (see (A.22)). Also, since $P''' > 0$, we must have $P(\lambda) > 0$ for λ large enough. Thus, there must be a $\lambda \in (1,\infty)$ with $P(\lambda) = 0$, and therefore at least one of the real roots of P is greater than 1. Since λ_2 is, by definition, the largest real root of P, this implies $\lambda_2 > 1$.

Note also that, from (A.24) and Assumption A4(ii), we have $P'(0) > 0$ for all $\phi \geq 1$, and therefore as noted above if $\lambda_{11}, \lambda_{12}$ are real, then they must both be strictly positive.

Next, by definition, we have $P(\lambda_2) = 0$. Totally differentiating this expression with respect to ϕ yields

$$\frac{d\lambda_2}{d\phi} = -\frac{1}{P'\left(\lambda_2\right)}\frac{\partial P\left(\lambda_2\right)}{\partial\phi}\,.$$

Since λ_2 is the largest real root of P and P is a cubic function with $P''' > 0$, we must have $P'(\lambda_2) > 0$. Further,

$$\frac{\partial P\left(\lambda_2\right)}{\partial\phi} = -\frac{\lambda_2-(1-\delta)}{\alpha_3}\lambda_2 < 0\,,$$

where the inequality follows since $\lambda_2 > 1 > 1-\delta$. Thus, we conclude that, for $\phi \geq 1$, $d\lambda_2/d\phi > 0$.

Next, from (A.21), the above implies that, for $\phi \geq 1$, we have $c > 0$ and $dc/d\phi < 0$. Since $c = \lambda_{11}\lambda_{12}$, and Proposition 9 establishes that $\lambda_{11}, \lambda_{12} \in (0,1)$ for the $\phi = 1$ case, we conclude that, on $\phi \geq 1$, c is a strictly positive and decreasing function of ϕ, that starts at a value below 1 for $\phi = 1$. Thus, $c \in (0,1)$ for all $\phi \geq 1$. This in turn implies that

$|\lambda_{11}| < 1$ for all $\phi \geq 1$. If $\lambda_{11}, \lambda_{12}$ are complex then they both have the same modulus, and therefore both are stable, in which case we have the desired saddle-path structure. Suppose instead $\lambda_{11}, \lambda_{12}$ are real. As noted above, this implies they are both positive. Further, since $|\lambda_{11}| < 1$, we have $\lambda_{11} \in (0,1)$. As already established, $P(1) < 0$ for this case, and from (A.23) we also have $P(0) < 0$. Thus, $P(\lambda)$ is negative for both $\lambda = 0$ and $\lambda = 1$, with a root (λ_{11}) in between those points. This necessarily implies that there must be two such roots, and therefore we also have $\lambda_{12} \in (0,1)$. We therefore again have two stable roots and one unstable one, i.e., the desired saddle-path structure. Thus, this structure persists for all $\phi \geq 1$, which completes the proof. □

Proposition 12

Note that ρ increasing from zero to one corresponds to ϕ decreasing from one to zero, with Assumption A3 ($\rho < 1$) being equivalent to $\phi > 0$. Note also that

$$\frac{\partial P(\lambda)}{\partial \phi} = \lambda \frac{1-\delta-\lambda}{\alpha_3} \begin{cases} < 0 & \text{, if } \lambda < 0 \\ > 0 & \text{, if } \lambda \in (0, 1-\delta) \\ < 0 & \text{, if } \lambda > 1-\delta \\ = 0 & \text{, if } \lambda \in \{0, 1-\delta\} \end{cases}, \tag{A.26}$$

$$\frac{\partial P(\lambda)}{\partial \alpha_1} = -\frac{\lambda}{\alpha_3} \begin{cases} > 0 & \text{, if } \lambda < 0 \\ = 0 & \text{, if } \lambda = 0 \\ < 0 & \text{, if } \lambda > 0 \end{cases},$$

$$\frac{\partial P'(\lambda)}{\partial \phi} = \frac{1-\delta-2\lambda}{\alpha_3} \begin{cases} > 0 & \text{, if } \lambda < (1-\delta)/2 \\ = 0 & \text{, if } \lambda = (1-\delta)/2 \\ < 0 & \text{, if } \lambda > (1-\delta)/2 \end{cases}.$$

We proceed by first formally stating a number of lemmas. Proofs of these lemmas are contained in Appendix A.3. In all cases, these proofs maintain Assumptions A4-A5, so that when $\phi = 1$ we have three real roots with $0 < \lambda_{11} \leq \lambda_{12} < 1 < \lambda_2$. For simplicity, we also take $\alpha_1 \neq 0$ and ignore knife-edge cases in the parameter space. It is nonetheless straightforward to extend the proofs to the degenerate case $\alpha_1 = 0$ and to these knife-edge cases.

Lemma A.1. *If $P(\lambda^*) > 0$ for some λ^*, then P has a real root $\lambda < \lambda^*$, and if in addition $\lambda^* < 0$ then it also has a real root in $(\lambda^*, 0)$. If $P(\lambda^*) < 0$ for some λ^*, then P has a real root $\lambda > \lambda^*$.*

Corollary 1. *$\lambda_2 > 0$ regardless of ϕ or any of the other parameters.*

It will often be useful to let $\lambda_j(\phi)$ denote the value of root $j \in \{11, 12, 2\}$ for a particular ϕ.

Lemma A.2. *Fix α_2, α_3, and δ, and suppose $\alpha_1 = \tilde{\alpha}_1$ and $\phi = \tilde{\phi}$. Then: (i) If $P(\lambda^*) = 0$ for some real $\lambda^* < 0$, then P has two real negative roots—one less than λ^*, the other in*

$(\lambda^*, 0)$—*for any combination of* $\alpha_1 \geq \tilde{\alpha}_1$ *and* $\phi \leq \tilde{\phi}$ *with* $(\alpha_1, \phi) \neq (\tilde{\alpha}_1, \tilde{\phi})$. *(ii) If* $P(\lambda) \leq 0$ *for all* $\lambda < 0$, *then* P *does not have any real negative roots for any combination of* $\alpha_1 \leq \tilde{\alpha}_1$ *and* $\phi \geq \tilde{\phi}$ *with* $(\alpha_1, \phi) \neq (\tilde{\alpha}_1, \tilde{\phi})$. *(iii) If* $P(\lambda) \leq 0$ *for all* $\lambda \in (0, 1-\delta)$, *then* P *does not have any real roots in* $(0, 1-\delta)$ *for any combination of* $\alpha_1 \geq \tilde{\alpha}_1$ *and* $\phi \leq \tilde{\phi}$ *with* $(\alpha_1, \phi) \neq (\tilde{\alpha}_1, \tilde{\phi})$ *. (iii) If* $P(\lambda^*) = 0$ *for some real* $\lambda^* \in (0, 1-\delta)$, *then* P *has at least one real root in* $(0, \lambda^*)$ *for all* $\phi > \tilde{\phi}$.

Lemma A.3. $\lambda = 0$ *is never a root of* P. *If* $\alpha_1 \neq 0$, $1 - \delta$ *is never a root of* P.

Lemma A.4. $P(\lambda) > 0$ *for* $\lambda > \lambda_2$, *and therefore* $P'(\lambda_2) \geq 0$ *(with strict inequality if and only if* $\lambda_{12} \neq \lambda_2$*).*

Lemma A.5. *Suppose* $\lambda_{11}, \lambda_{12}$ *are real. Then* $P(\lambda) < 0$ *for* $\lambda < \lambda_{11}$, $P'(\lambda_{11}) \geq 0$ *(with strict inequality if and only if* $\lambda_{11} < \lambda_{12}$*), and* $P'(\lambda_{12}) \leq 0$ *(with strict inequality if and only if all three roots are distinct). Further,* $P(\lambda) > 0$ *on* $(\lambda_{11}, \lambda_{12})$, *and* $P(\lambda) < 0$ *on* $(\lambda_{12}, \lambda_2)$.

Lemma A.6. *If* $\alpha_1 > 0$ *then* $\lambda_2 > 1 - \delta$. *If* $\alpha_1 = 0$ *then* $\lambda_2 \geq 1 - \delta$. *If* $\alpha_1 < 0$ *then there is always a real root in* $(0, 1-\delta)$.

Lemma A.7. *Suppose the roots of* P *are all real. Then* $\tilde{\lambda}$ *is a multiple root of* P *if and only if* $P(\tilde{\lambda}) = P'(\tilde{\lambda}) = 0$. *If* $\lambda_{11} = \lambda_{12} = \tilde{\lambda} < \lambda_2$, *then* $P''(\tilde{\lambda}) < 0$ *(so that* $P(\lambda)$ *achieves a local maximum at* $\lambda = \tilde{\lambda}$*). If* $\lambda_{11} < \lambda_{12} = \lambda_2 = \tilde{\lambda}$, *then* $P''(\tilde{\lambda}) > 0$ *(so that* $P(\lambda)$ *achieves a local minimum at* $\lambda = \tilde{\lambda}$*).*

Lemma A.8. *(i) If* λ_{11} *is not a multiple root, then* $\lambda'_{11}(\phi) > 0$ *if* $\lambda_{11}(\phi) \in (-\infty, 0) \cup (1-\delta, \infty)$, *while* $\lambda'_{11}(\phi) < 0$ *if* $\lambda_{11}(\phi) \in (0, 1-\delta)$; *(ii) If* λ_{12} *is not a multiple root, then* $\lambda'_{12}(\phi) < 0$ *if* $\lambda_{12}(\phi) \in (-\infty, 0) \cup (1-\delta, \infty)$, *while* $\lambda'_{12}(\phi) > 0$ *if* $\lambda_{12}(\phi) \in (0, 1-\delta)$; *and (iii) If* λ_2 *is not a multiple root, then* $\lambda'_2(\phi) > 0$ *if* $\lambda_2(\phi) > 1 - \delta$, *while* $\lambda'_2(\phi) < 0$ *if* $\lambda_2(\phi) \in (0, 1-\delta)$.

Lemma A.9. *There exists a* $\bar{\phi} \in (-\infty, 1)$ *such that* $\lambda_{11}, \lambda_{12}$ *are real and negative if and only if* $\phi \leq \bar{\phi}$, *with* $\lambda_{11}(\bar{\phi}) = \lambda_{12}(\bar{\phi})$. *Further,* $\lambda'_{11}(\phi) > 0$ *and* $\lambda'_{12}(\phi) < 0$ *on* $\phi < \bar{\phi}$, *and there exists a unique value of* ϕ *for which* $P(-1) = 0$. *Lastly, for* $\phi > \bar{\phi}$ *small enough we have* $\lambda_{11}, \lambda_{12}$ *complex.*

Corollary 2. *There is exactly one bifurcation involving a root equal to* -1 *as* ϕ *decreases from 1 to* $-\infty$.

Lemma A.10. *Suppose there is some* ϕ_1 *such that the roots are all real for* $\phi = \phi_1$, *with* $\lambda_{12}(\phi_1) > 1-\delta$. *Then there exists a* $\phi_2 \leq \phi_1$ *such that (i) the roots are all real for* $\phi \in [\phi_2, \phi_1]$; *(ii)* $\lambda_{11}(\phi) = \lambda_{12}(\phi)$ *on* $\phi \in [\phi_2, \phi_1)$ *only if* $\phi = \phi_2$; *(iii)* $\lambda_{12}(\phi) = \lambda_2(\phi)$ *on* $\phi \in [\phi_2, \phi_1]$ *if and only if* $\phi = \phi_2$; *(iv) as* ϕ *decreases from* ϕ_1 *to* ϕ_2, λ_2 *decreases, while* λ_{12} *increases; and (v) for all* $\phi \in [\phi_2, \phi_1)$, *we have* $\lambda_{11}(\phi) \in (1-\delta, \lambda_{11}(\phi_1))$ *if* $\lambda_{11}(\phi_1) > 1 - \delta$, *and* $\lambda_{11}(\phi) \in (\lambda_{11}(\phi_1), 1-\delta)$ *if* $\lambda_{11}(\phi_1) < 1 - \delta$.

Lemma A.11. *Suppose there is some* ϕ_2 *such that the roots are all real for* $\phi = \phi_2$, *with* $1-\delta < \lambda_{11}(\phi_2) < \lambda_{12}(\phi_2) = \lambda_2(\phi_2)$. *Then, for* $\phi \in (\bar{\phi}, \phi_2)$ *(where* $\bar{\phi}$ *is the value from Lemma A.9 at which* $\lambda_{11}(\bar{\phi}) = \lambda_{12}(\bar{\phi}) < 0$*),* $\lambda_{11}, \lambda_{12}$ *are complex.*

Lemma A.12. *Suppose $1-\delta < \lambda_{12}(\phi_0) < 1 < \lambda_2(\phi_0)$ for some $\phi_0 \le 1$. Then the first bifurcation that occurs as ϕ decreases from ϕ_0 is a fold bifurcation.*

Lemma A.13. *Suppose $0 < \lambda_{11}(1) \le \lambda_{12}(1) < 1-\delta$ and $\lambda_2(1) > 1$. Then as ϕ decrease from 1, λ_{11} increases and λ_{12} decreases until they collide at a value in $(0, 1-\delta)$ when $\phi = \phi_0 \le 1$, and then become complex as ϕ decreases through ϕ_0. As ϕ continues to decrease, the roots next become real at a value $\tilde{\lambda}$ when $\phi = \phi_1 < \phi_0$. Either $\tilde{\lambda} < 0$ (in which case $\phi_1 = \bar{\phi}$ from Lemma A.9) or $\tilde{\lambda} > 1-\delta$.*

Corollary 3. *Starting from a configuration with $0 < \lambda_{11}(1) \le \lambda_{12}(1) < 1 < \lambda_2(1)$, as ϕ decreases from 1 there are at most two disjoint intervals of ϕ for which $\lambda_{11}, \lambda_{12}$ are complex.*

Lemma A.14. *Suppose $\alpha_1 > 0$ and let ϕ_0, ϕ_1, $\phi_0 > \phi_1$, be such that for $\phi \in \{\phi_1, \phi_0\}$ there are three real roots, while for $\phi \in (\phi_1, \phi_0)$ the roots are complex. Then $\partial|\lambda_{12}|/\partial\phi < 0$ on $\phi \in (\phi_1, \phi_0)$, so that $|\lambda_{12}(\phi_1)| > |\lambda_{12}(\phi_0)|$.*

Lemma A.15. *As ϕ decreases from 1, a hysteresis-inducing fold occurs if and only if $\alpha_1 < \alpha_1^{**} \equiv (\alpha_2 - \alpha_3)\delta^2$. Further, if in fact $\alpha_1 < \alpha_1^{**}$, then this hysteresis-inducing fold is the first bifurcation.*

We turn now to proving the Proposition. To see part (iv), note that a fold bifurcation requires $P(1) = 0$. Since $P(1) < 0$ for $\phi = 1$ and $\partial P(1)/\partial\phi < 0$, we see that $P(1) < 0$ for all $\phi \in [0, 1)$ if and only if $\alpha_1 \ge \delta(\alpha_2 + \alpha_3)$. Part (iv) then follows immediately. Part (iii) is simply Lemma A.15.

Thus, assume henceforth that $\alpha_1 > \alpha_1^{**}$, so that if a fold occurs it must be an indeterminacy-inducing one. Given α_2 and α_3, define

$$\alpha_1^* \equiv (\alpha_2 - \alpha_3)(2-\delta)^2\,,$$
$$\phi^* \equiv \alpha_2(1-\delta) - \alpha_3(3-\delta) < 1\,,$$
$$\lambda_2^* \equiv (1-\delta)\frac{\alpha_2}{\alpha_3} > 0\,.$$

It can be verified that if $\alpha_1 = \alpha_1^*$ and $\phi = \phi^*$, then we have $P(\lambda_2^*) = P(-1) = P'(-1) = 0$, so that the roots are given by $\lambda_{11} = \lambda_{12} = -1$ and $\lambda_2 = \lambda_2^* > 0$. By Lemma A.5, this implies that $P(\lambda) \le 0$ on $\lambda < 0$, with the inequality strict for $\lambda \ne -1$.

Next, suppose $\phi = \phi^*$. Since for $\alpha_1 = \alpha_1^*$ we have $P(-1) = 0$, Lemma A.2(i) implies that if $\alpha_1 > \alpha_1^*$ then P will have two negative roots, one greater than and one less than -1, i.e., $\lambda_{11} < -1 < \lambda_{12} < 0$ for $\alpha_1 > \alpha_1^*$. Meanwhile, since for $\alpha_1 = \alpha_1^*$ we have $P(\lambda) \le 0$ on $\lambda < 1$, Lemma A.2(ii) implies that if $\alpha_1 < \alpha_1^*$ then P does not have any real negative roots. Since the roots of P are continuous in α_1, and $\lambda_{11} = \lambda_{12} = -1$ for $\alpha_1 = \alpha_1^*$, $\lambda_{11}, \lambda_{12}$ must initially become complex as α_1 decreases from α_1^*.

Now,

$$\frac{\partial P(-1)}{\partial \alpha_1} = \frac{1}{\alpha_3} > 0\,, \qquad \frac{\partial P(-1)}{\partial \phi} = -\frac{2-\delta}{\alpha_3} < 0\,.$$

Thus, beginning from $\alpha_1 = \alpha_1^*$ and $\phi = \phi^*$ (so that $P(-1) = 0$), then as α_1 increases or decreases we can keep $P(-1) = 0$ by moving ϕ in the same direction as α_1. Specifically, if we set

$$\phi = \phi^*(\alpha_1) \equiv \phi^* + \frac{1}{2-\delta}(\alpha_1 - \alpha_1^*) = \frac{1}{2-\delta}\alpha_1 - (\alpha_2 + \alpha_3)\,.$$

then $P(-1)=0$ for all α_1. Thus, at least one root is equal to -1 for all α_1, and since $\lambda_2>0$ (Corollary 1), it must be one of $\lambda_{11},\lambda_{12}$ (or both). Next, if we substitute $\phi=\phi^*(\alpha_1)$ into P, we may then obtain that

$$\frac{\partial P'(-1)}{\partial \alpha_1}=\frac{1}{(2-\delta)\,\alpha_3}>0\,.$$

Since with $\phi=\phi^*(\alpha_1)$ we have $P'(-1)=0$ when $\alpha_1=\alpha_1^*$, for $\alpha_1>\alpha_1^*$ we must have $P'(-1)>0$, and therefore from Lemma A.5 we must have $\lambda_{11}=-1<\lambda_{12}<0$. Conversely, we have $P'(-1)<0$ for $\alpha_1<\alpha_1^*$, and therefore Lemma A.5 implies $\lambda_{11}<-1=\lambda_{12}$. We therefore see that, for $\alpha_1\neq\alpha_1^*$, a flip bifurcation occurs at $\phi=\phi^*(\alpha_1)$.[18]

Now, fix any $\alpha_1\neq\alpha_1^*$, so that when $\phi=\phi^*(\alpha_1)$ we have $\lambda_{11}(\phi^*(\alpha_1))<\lambda_{12}(\phi^*(\alpha_1))<0$. Note that this necessarily implies that $\phi^*(\alpha_1)<\bar{\phi}$, where $\bar{\phi}$ is as defined in Lemma A.9. Thus, as ϕ increases from $\phi^*(\alpha_1)$, by Lemma A.9 λ_{11} increases and λ_{12} decreases, until they collide when $\phi=\bar{\phi}$, and then become complex for ϕ just above $\bar{\phi}$.

Suppose $\alpha_1>\alpha_1^*$, so that $-1<\lambda_{11}(\phi^*(\alpha_1))<\lambda_{12}(\phi^*(\alpha_1))<0$. Then we have $-1<\lambda_{11}\leq\lambda_{12}<0$ for $\phi\in(\phi^*(\alpha_1),\bar{\phi}]$, so that the unique flip bifurcation occurs as ϕ decreases through $\phi^*(\alpha_1)$, and this involves λ_{11} decreasing through -1. Since for $\phi\leq\bar{\phi}$ we have $\lambda_{11},\lambda_{12}$ real and negative, that unique flip bifurcation is also the only bifurcation involving $\lambda_{11},\lambda_{12}$ that can occur for $\phi\leq\bar{\phi}$. Thus, if no bifurcation has occurred for $\phi\in(\bar{\phi},1)$, the first bifurcation must be a flip.

Case 1: $\lambda_{12}(1)\in(1-\delta,1)$. From Lemma A.12 the first bifurcation is a fold, and therefore as noted above it must be an indeterminacy-inducing one.

Case 2: $0<\lambda_{11}(1),\lambda_{12}(1)<1-\delta$. Note that this requires $\alpha_1>0$. From Lemma A.13, we know that as ϕ decreases from 1, the first transition is that $\lambda_{11},\lambda_{12}$ becomes complex as ϕ decreases through $\phi_0<1$, with $\lambda_{12}(\phi_0)\in(0,1-\delta)$, and the second transition is that they become real at a value $\tilde{\lambda}$ that is either negative or greater than $1-\delta$ when $\phi=\phi_1<\phi_0$. If $\tilde{\lambda}<0$, then $\phi_1=\bar{\phi}$, and since $\alpha_1>\alpha_1^*$, as noted above we must have $\tilde{\lambda}\in(-1,0)$. By Lemma A.14, $|\lambda_{12}|$ strictly increases as ϕ decreases from ϕ_0 to ϕ_1, and therefore we also have $|\lambda_{12}(\phi)|<1$ for all $\phi\in(\phi_1,\phi_0)$. If $\lambda_2(\phi_1)<1$ then the first bifurcation clearly must have been an indeterminacy-inducing fold at some $\phi\in(\phi_1,\phi_0)$, while if $\lambda_2(\phi_1)>1$ then no bifurcation occurred on (ϕ_1,ϕ_0), and thus the first bifurcation must be either the flip at $\phi=\phi^*(\alpha_1)<\phi_1$ or an indeterminacy-inducing fold at some $\phi\in(\phi^*(\alpha_1),\phi_1)$.

Suppose instead $\tilde{\lambda}\in(1-\delta,1)$. By Lemma A.14, we again have $|\lambda_{12}|$ strictly increasing as ϕ decreases from ϕ_0 to ϕ_1, and thus $|\lambda_{12}(\phi)|<1$ for all $\phi\in(\phi_1,\phi_0)$. If $\lambda_2(\phi_1)<1$ then the first bifurcation clearly must have been an indeterminacy-inducing fold at some $\phi\in(\phi_1,\phi_0)$. If instead $\lambda_2(\phi_1)>1$, then no bifurcation occurred on (ϕ_1,ϕ_0), and further $\lambda_{12}(\phi_1)<1<\lambda_2(\phi_1)$, so that by Lemma A.12 the first bifurcation must be an indeterminacy-inducing fold.

Finally, suppose $\tilde{\lambda}>1$. By Lemma A.10, there is a $\phi_2\leq\phi_1$ such that $\lambda_{12}(\phi_2)=\lambda_2(\phi_2)=\hat{\lambda}>\lambda_{12}(\phi_1)>1$, and by Lemma A.11 as ϕ decreases below ϕ_2 the double root at $\hat{\lambda}$ becomes complex and remains so until $\phi=\bar{\phi}$. By Lemma A.14, we thus have $\lambda_{12}(\bar{\phi})<-\lambda_{12}(\phi_2)<-1$. But this contradicts the fact that, since $\alpha_1>\alpha_1^*$, we must have $\lambda_{12}(\bar{\phi})\in(-1,0)$. Thus, we

[18]For the knife-edge case $\alpha_1=\alpha_1^*$, there is also a bifurcation at $\phi=\phi^*$, but this is a more complex bifurcation involving a pair of negative roots colliding at -1 and then immediately becoming complex. Technically this is neither a flip nor a Hopf bifurcation.

cannot in fact have $\tilde{\lambda} > 1$ when $\alpha_1 > \alpha_1^*$.

Suppose now that $\alpha_1 < \alpha_1^*$, so that $\lambda_{11}(\phi^*(\alpha_1)) < -1 = \lambda_{12}(\phi^*(\alpha_1))$. Then we have $\lambda_{11} \leq \lambda_{12} < -1$ for $\phi \in (\phi^*(\alpha_1), \bar{\phi}]$, which implies that $\lambda_{11}, \lambda_{12}$ were already unstable at $\phi = \bar{\phi}$. Thus, the flip bifurcation experienced as ϕ decreases through $\phi^*(\alpha_1)$ cannot be the first bifurcation to occur as ϕ decreases from 1. Since at most one flip bifurcation can occur (Corollary 2), the prior bifurcation must have been either a Hopf or a fold.

In terms of α_2, the conditions described above are:

$$\alpha_1 > \alpha_1^* \Leftrightarrow \alpha_2 < \alpha_2^* \equiv \frac{\alpha_1}{(2-\delta)^2} + \alpha_3$$

$$\alpha_1 > \alpha_1^{**} \Leftrightarrow \alpha_2 < \alpha_2^{**} \equiv \frac{\alpha_1}{\delta^2} + \alpha_3 > \alpha_2^*$$

$$\alpha_2^* < \alpha_2 < \alpha_2^{**}$$

We have therefore so far confirmed part (iii) and (iv) of the Proposition, along with the necessary conditions stated in parts (i) and (ii) for the first bifurcation to be a flip, Hopf, or indeterminacy-inducing fold.

For part (i), it remains to confirm that the condition $\alpha_2 < \alpha_2^*$ is also sufficient for the first bifurcation to be a flip. Since the only other possibility for this case is that an (indeterminacy-inducing) fold is the first bifurcation, we need only rule this possibility out. Thus, suppose $\alpha_2 < \alpha_2^*$. Note that, even if it is not the first bifurcation, a flip necessarily occurs at $\phi = \phi^*(\alpha_1)$. This value satisfies Assumption A3 (i.e., $\phi^*(\alpha_1) > 0$) if and only if

$$\alpha_1 > (\alpha_2 + \alpha_3)(2-\delta)\ . \tag{A.27}$$

Now, suppose a fold bifurcation occurred at some $\phi = \tilde{\phi} \in (\phi^*(\alpha_1), 1)$, i.e., we had $P(1) = 0$ for this value of ϕ. Since $\partial P(1)/\partial\phi < 0$, this implies that for $\phi = \phi^*(\alpha_1) < \tilde{\phi}$ we must have

$$P(1) = 2\frac{(\alpha_2 + \alpha_3)\,\delta - \frac{1}{2-\delta}\alpha_1}{\alpha_3} > 0\,,$$

which is in turn equivalent to

$$\alpha_1 > (\alpha_2 + \alpha_3)(2-\delta)\,\delta\,. \tag{A.28}$$

Conditions (A.27) and (A.28) clearly cannot both hold, and therefore it is not possible for a fold bifurcation to occur. We therefore conclude that when $\alpha_2 < \alpha_2^*$ the only possibility is a flip bifurcation, which completes the proof of part (i).

For part (ii), we have already established that if $\alpha_2^* < \alpha_2 < \alpha_2^{**}$ then the only possibilities for the first bifurcation are a Hopf or an indeterminacy-inducing fold. The last thing we need to establish is that it is the former if and only if $\alpha_2 > \alpha_3/(1-\delta)$.

To see the "only if" part, suppose a Hopf is the first bifurcation, and that it occurs at $\phi = \tilde{\phi}$. Since $c = \lambda_{11}(\tilde{\phi})\lambda_{12}(\tilde{\phi}) = 1$, from (A.21) we must have $\lambda_2(\tilde{\phi}) = (1-\delta)\alpha_2/\alpha_3$. Since this is the first bifurcation by hypothesis, we must have $\lambda_2(\tilde{\phi}) > 1$, which is the case if and only if $\alpha_2 > \alpha_3/(1-\delta)$. This confirms the "only if" part.

To see the "if" part, suppose instead that the first bifurcation is an indeterminacy-inducing fold, which happens at some $\phi = \tilde{\phi}$, so that $\lambda_2(\tilde{\phi}) = 1$. Since this fold is the first bifurcation by hypothesis, we must have $|\lambda_{1j}(\tilde{\phi})| < 1$ for $j = 1, 2$, and therefore $c = \lambda_{11}(\tilde{\phi})\lambda_{12}(\tilde{\phi}) < 1$. From (A.21), we see that for $\phi = \tilde{\phi}$ we have $c = (1-\delta)\alpha_2/\alpha_3$, which is less than 1 if and only if $\alpha_2 < \alpha_3/(1-\delta)$, which confirms the "if" part. □

A.3 Lemmas for Proof of Proposition 12

A.3.1 Proof of Lemma A.1

Since $P''' > 0$, we must always have $P(\lambda) < 0$ for λ sufficiently negative, and $P(\lambda) > 0$ for λ sufficiently positive. Furthermore, $P(0) < 0$. Each element of the lemma then follows immediately by the intermediate value theorem. □

A.3.2 Proof of Corollary 1

Since $P(0) = -(1-\delta)\alpha_2/\alpha_3 < 0$, the result follows immediately from Lemma A.1. □

A.3.3 Proof of Lemma A.2

Note that, for $\lambda < 0$, we have $\partial P(\lambda)/\partial\alpha_1 > 0$ and $\partial P(\lambda)/\partial\phi < 0$, i.e., $P(\lambda)$ is strictly increasing in α_1 and strictly decreasing in ϕ.

To see (i), suppose $P(\lambda^*) = 0$ for some real $\lambda^* < 0$ when $\alpha_1 = \tilde{\alpha}_1$ and $\phi = \tilde{\phi}$. Since $P(\lambda^*)$ is increasing in α_1/decreasing in ϕ, for any $\alpha_1 \geq \tilde{\alpha}_1$ and $\phi \leq \tilde{\phi}$ with at least one of the inequalities strict we will have $P(\lambda^*) > 0$. The rest of (i) then follows directly from Lemma A.1.

To see (ii), suppose P does not have any real negative roots for $\alpha_1 = \tilde{\alpha}_1$ and $\phi = \tilde{\phi}$, i.e., $P(\lambda) < 0$ for all $\lambda < 0$. Since $P(\lambda^*)$ is increasing in α_1/decreasing in ϕ, for any $\alpha_1 \leq \tilde{\alpha}_1$ and $\phi \geq \tilde{\phi}$ with at least one of the inequalities strict we must also have $P(\lambda) < 0$ for all $\lambda < 0$, and therefore no real negative roots.

After noting that $\partial P(\lambda)/\partial\alpha_1 < 0$ and $\partial P(\lambda)/\partial\phi > 0$ for $\lambda \in (0, 1-\delta)$, the proof of (iii) is symmetric to (ii), just with α_1 and ϕ moving in the opposite directions from that case.

To see (iv), suppose $P(\lambda^*) = 0$ for some real $\lambda^* \in (0, 1-\delta)$ when $\phi = \tilde{\phi}$. Since $\partial P(\lambda)/\partial\phi > 0$ for $\lambda \in (0, 1-\delta)$, we have $P(\lambda^*) > 0$ for $\phi > \tilde{\phi}$, and since $P(0) < 0$, by the intermediate value theorem there must be a root in $(0, \lambda^*)$. □

A.3.4 Proof of Lemma A.3

The lemma follows immediately from the fact that $P(0) < 0$ for all parameter values, while

$$P(1-\delta) = -\frac{\alpha_1(1-\delta)}{\alpha_3}$$

which is non-zero as long as $\alpha_1 \neq 0$. □

A.3.5 Proof of Lemma A.4

Suppose $\lambda > \lambda_2$ is such that $P(\lambda) \leq 0$. If $P(\lambda) = 0$ then $\lambda > \lambda_2$ is a real root of P, while if $P(\lambda) < 0$ then by Lemma A.1 P has a real root greater than $\lambda > \lambda_2$. Either way this implies the existence of a real root greater than λ_2, which contradicts the definition of λ_2 as the largest real root. Thus, $P(\lambda) > 0$ for $\lambda > \lambda_2$, and therefore we must have $P'(\lambda_2) \geq 0$. Further, by Lemma A.7, $P'(\lambda_2) = 0$ if and only if P has a multiple root at $\lambda = \lambda_2$, which is in turn the case if and only if $\lambda_{12} = \lambda_2$. Thus, the inequality is strict as long as $\lambda_{12} \neq \lambda_2$. $\square$

A.3.6 Proof of Lemma A.5

Suppose $\lambda < \lambda_{11}$ is such that $P(\lambda) \geq 0$. If $P(\lambda) = 0$ then $\lambda < \lambda_{11}$ is a real root of P, while if $P(\lambda) > 0$ then by Lemma A.1 P has a real root less than $\lambda < \lambda_{11}$. Either way this implies the existence of a real root less than λ_{11}, which contradicts the definition of λ_{11} as the smallest real root. Thus, $P(\lambda) < 0$ for $\lambda < \lambda_{11}$, and therefore we must have $P'(\lambda_{11}) \geq 0$. Further, by Lemma A.7, $P'(\lambda_{11}) = 0$ if and only if P has a multiple root at $\lambda = \lambda_{11}$, which is in turn the case if and only if $\lambda_{11} = \lambda_{12}$. Thus, $P'(\lambda_{11}) > 0$ as long as $\lambda_{11} < \lambda_{12}$. If this is the case, then $P(\lambda) > 0$ for λ just above λ_{11}, and therefore for $\lambda \in (\lambda_{11}, \lambda_{12})$ we must have $P(\lambda) > 0$. This in turn implies that $P(\lambda) > 0$ for λ just below λ_{12}, and thus $P'(\lambda_{12}) \leq 0$. By Lemma A.7, $P'(\lambda_{12}) = 0$ if and only if $\lambda = \lambda_{12}$ is a multiple root, which in turn occurs if and only if either $\lambda_{12} = \lambda_{11}$ or $\lambda_{12} = \lambda_2$ (or both). Thus, $P'(\lambda_{12}) < 0$ if and only if all three roots are distinct.

Lastly, suppose $\lambda_{12} < \lambda_2$. Then Lemma A.4 implies that $P'(\lambda_2) > 0$, and therefore for λ just below λ_2 we have $P(\lambda) < 0$. Thus, we must have $P(\lambda) < 0$ on $(\lambda_{12}, \lambda_2)$. $\square$

A.3.7 Proof of Lemma A.6

If $\alpha_1 > 0$, then since $P(1-\delta) = -(1-\delta)\alpha_1/\alpha_3$, we have $P(1-\delta) < 0$, and therefore by Lemma A.1 we must have $\lambda_2 > 1-\delta$. If $\alpha_1 = 0$, we have $P(1-\delta) = 0$, and therefore $1-\delta$ is always a root of P, in which case $\lambda_2 \geq 0$. Finally, if $\alpha_1 < 0$, then $P(1-\delta) > 0$, and since $P(0) < 0$, by the intermediate value theorem there must be a root in $(0, 1-\delta)$. $\square$

A.3.8 Proof of Lemma A.7

By definition, we may write $P(\lambda) = (\lambda - \lambda_{11})(\lambda - \lambda_{12})(\lambda - \lambda_2)$, and therefore

$$P'\left(\tilde{\lambda}\right) = \left(\tilde{\lambda} - \lambda_{12}\right)\left(\tilde{\lambda} - \lambda_2\right) + \left(\tilde{\lambda} - \lambda_{11}\right)\left(\tilde{\lambda} - \lambda_2\right) + \left(\tilde{\lambda} - \lambda_{11}\right)\left(\tilde{\lambda} - \lambda_{12}\right).$$

Suppose $P(\tilde{\lambda}) = P'(\tilde{\lambda}) = 0$. Then clearly $\tilde{\lambda}$ is a root. Further, to have $P'(\tilde{\lambda}) = 0$, at least one of the two factors in each of the terms above must be zero. If all three roots are distinct, this is not possible. Thus, $\tilde{\lambda}$ must be a multiple root.

Next, suppose $\tilde{\lambda}$ is a double (or triple) real root of P, so that we may write $P(\lambda) = (\lambda - \tilde{\lambda})^2(\lambda - \bar{\lambda})$ for some real $\bar{\lambda}$. Thus, clearly $P'(\tilde{\lambda}) = 0$, while $P''(\tilde{\lambda}) = 2(\tilde{\lambda} - \bar{\lambda})$. If $\lambda_{11} = \lambda_{12} = \tilde{\lambda} < \lambda_2$ then $\bar{\lambda} = \lambda_2 > \tilde{\lambda}$, in which case $P''(\tilde{\lambda}) < 0$, while if $\lambda_{11} < \lambda_{12} = \lambda_2 = \tilde{\lambda}$ then $\bar{\lambda} = \lambda_{11} < \tilde{\lambda}$, in which case $P''(\tilde{\lambda}) > 0$. $\square$

A.3.9 Proof of Lemma A.8

By Lemmas A.4 and A.5, we have $P'(\lambda_{11}), P'(\lambda_2) > 0$ and $P'(\lambda_{12}) < 0$ as long as each is respectively not a multiple root. Since, for $j \in \{11, 12, 2\}$, we have $\lambda_j'(\phi) = -[\partial P(\lambda_j)/\partial\phi]/P'(\lambda_j)$, the results then follow from (A.26). $\square$

A.3.10 Proof of Lemma A.9

Since for $\phi = 1$ all roots are strictly positive, for ϕ in a neighborhood of 1 we must have $P(\lambda) < 0$ for all $\lambda < 0$. Fix $\lambda < 0$. Since $\partial P(\lambda)/\partial\phi$ is negative and independent of ϕ, $P(\lambda)$ increases without bound as ϕ decreases from 1. Thus, we must eventually have $P(\lambda) > 0$, and therefore by Lemma A.1 we must eventually have two real negative roots. Let $\bar{\phi} < 1$ be the first value of ϕ where a real negative root appears as ϕ decreases from 1. By Lemma A.3, a real root cannot transition directly from positive to negative, and thus when a real negative root first appears it must do so as a pair of complex roots becomes real, in which case we have double root for $\phi = \bar{\phi}$: $\lambda_{11}(\bar{\phi}) = \lambda_{12}(\bar{\phi})$. By definition of $\bar{\phi}$, neither root is real and negative if $\phi > \bar{\phi}$, while by Lemma A.2 both roots are necessarily also negative for $\phi < \bar{\phi}$. Further, Lemma A.8 applies for $\phi < \bar{\phi}$, so that $\lambda_{11}'(\phi) > 0$ and $\lambda_{12}'(\phi) < 0$. Next, since $P(-1) < 0$ for $\phi = 1$, and since $\partial P(-1)/\partial\phi = -(2-\delta)/\alpha_3 < 0$, $P(-1)$ is a strictly decreasing linear function of ϕ, and therefore $P(-1) = 0$ for exactly one value of ϕ. Finally, the fact that $\lambda_{11}, \lambda_{12}$ are a complex pair for ϕ just above $\bar{\phi}$ follows immediately from the continuity of the roots in ϕ[19] and the fact that $\lambda_{11}(\bar{\phi}) = \lambda_{12}(\bar{\phi}) < 0$. $\square$

A.3.11 Proof of Corollary 2

Since such a bifurcation necessarily involves $P(-1) = 0$, this follows directly from the last property of Lemma A.9. $\square$

A.3.12 Proof of Lemma A.10

Note that $\lambda_2(\phi_1) \geq \lambda_{12}(\phi_1) > 1-\delta$. If $\lambda_{12}(\phi_1) = \lambda_2(\phi_1)$, then $\phi_2 = \phi_1$, and the lemma follows trivially. Thus, suppose henceforth that $\lambda_2(\phi_1) > \lambda_{12}(\phi_1) = \tilde{\lambda} > 1-\delta$. Suppose $\lambda_{11}(\phi_1) = \tilde{\lambda}$ as well, which implies that $P(1-\delta) < 0$ for all ϕ. Then by Lemma A.7, for $\phi = \phi_1$, we have $P'(\tilde{\lambda}) = 0$, with P achieving a local maximum at $\tilde{\lambda}$, and by Lemma A.5 we have $P(\lambda) < 0$ for $\lambda < \tilde{\lambda}$. Since $\tilde{\lambda} > 1-\delta$, we have $\partial P(\tilde{\lambda})/\partial\phi < 0$, and since $P(\tilde{\lambda}) = 0$ for $\phi = \phi_1$, it follows that for $\phi < \phi_1$ we have $P(\tilde{\lambda}) > 0$, and since $P(1-\delta) < 0$, by the intermediate value theorem P has a root in $(1-\delta, \tilde{\lambda})$ for all $\phi < \phi_1$. Meanwhile, since $\tilde{\lambda} < \lambda_2(\phi_1)$, Lemma A.5 implies that for $\phi = \phi_1$ we have $P(\lambda) < 0$ for all $\lambda \in (\tilde{\lambda}, \lambda_2(\phi_1))$. Thus, for ϕ just below ϕ_1 there must still be a $\bar{\lambda} \in (\tilde{\lambda}, \lambda_2(\phi_1))$ with $P(\bar{\lambda}) < 0$, and therefore by the intermediate value

[19] The roots are continuous in the sense that if $\tilde{\lambda}$ is a root when $\phi = \tilde{\phi}$, then for any $\epsilon > 0$, there exists a $\zeta > 0$ such that P has a root λ satisfying $|\lambda - \tilde{\lambda}| < \epsilon$ whenever $|\phi - \tilde{\phi}| < \zeta$. Note that this does not necessarily imply that $\lambda_j(\phi)$ is always a continuous function for $j \in \{11, 12, 2\}$. Specifically, $\lambda_{11}(\phi)$ and $\lambda_2(\phi)$ will not be continuous functions at $\phi = \tilde{\phi}$ if $\lambda_{11}(\tilde{\phi}) < \lambda_{12}(\tilde{\phi}) = \lambda_2(\tilde{\phi}) = \tilde{\lambda} \neq 1-\delta$: because the double root at $\tilde{\lambda}$ becomes complex in a continuous way for ϕ on one side of $\tilde{\phi}$, and by definition λ_2 is never complex, $\lambda_{11}(\phi)$ and $\lambda_2(\phi)$ effectively swap roles at $\tilde{\phi}$, so that at root near $\tilde{\lambda}$ is called λ_2 on one side of $\tilde{\phi}$ and λ_{11} on the other. This is just an artifact of the way we have defined the labels $\lambda_{11}, \lambda_{12}, \lambda_2$, however. The roots themselves are nonetheless continuous in ϕ everywhere.

theorem there must be a real root in $(\tilde{\lambda}, \bar{\lambda})$, and by Lemma A.1 another real root that is greater than $\bar{\lambda}$. Thus, for ϕ just below ϕ_1 we have $\lambda_2 > \lambda_{12} > \lambda_{11} \in (1-\delta, \lambda_{11}(\phi_1))$. In the case where instead $\lambda_{11}(\phi_1) \in (1-\delta, \tilde{\lambda})$, the only difference is that we also have this structure even when $\phi = \phi_1$. The hypotheses of Lemma A.8 then hold, so that as ϕ decreases from ϕ_1, λ_{11} and λ_2 decrease, while λ_{12} increases. The hypotheses of Lemma A.8 clearly continue to then hold up until the point where λ_{12} and λ_2 collide. Letting ϕ_2 be the value at which this happens, we have confirmed all parts of the lemma for the case where $\lambda_{11}(\phi_1) > 1-\delta$. In the case where $0 < \lambda_{11}(\phi_1) < 1-\delta < \lambda_{12}(\phi_1) < \lambda_2(\phi_1)$, everything is identical to the $1-\delta < \lambda_{11}(\phi_1) < \lambda_{12}(\phi_1) < \lambda_2(\phi_1)$ case, except by Lemma A.8 λ_{11} increases as ϕ decreases. However, since a real root cannot pass through $1-\delta$, we have $\lambda_{11} < 1-\delta$ on $[\phi_2, \phi_1]$, which in turn implies that λ_{11} cannot collide with λ_{12} before the latter collides with λ_2, This completes the proof. □

A.3.13 Proof of Lemma A.11

The given structure for $\phi = \phi_2$ implies that $P(1-\delta) < 0$, and therefore $\alpha_1 > 0$, $\lambda_2(\phi) > 1-\delta$ for all ϕ, and that $1-\delta$ can never be a root of P for any ϕ. Next, since $\lambda_{12}(\phi_2) = \lambda_2(\phi_2) = \tilde{\lambda}$ is a double root when $\phi = \phi_2$ with $\lambda_{11}(\phi_2) < \tilde{\lambda}$ the other root, for $\phi = \phi_2$ we have $P(\lambda) < 0$ for $\lambda < \lambda_{11}$ (Lemma A.5), $P(\lambda) > 0$ for $\lambda \in (\lambda_{11}, \tilde{\lambda}) \cup (\tilde{\lambda}, \infty)$ (Lemmas A.4 and A.5), and the double root $\tilde{\lambda}$ is a local minimizer of P (Lemma A.7). Since $\lambda_{11}(\phi_2) > 1-\delta$, we have $P(\lambda) < 0$ on $(0, 1-\delta)$ for $\phi = \phi_2$, and therefore from Lemma A.2 there cannot be any roots in $(0, 1-\delta)$ for $\phi < \phi_2$ either. Since we have already ruled out roots at 0 and $1-\delta$, we conclude that there cannot be any roots in $[0, 1-\delta]$ for $\phi \leq \phi_2$.

Next, since, for $\phi = \phi_2$, $P(\lambda) \geq 0$ on $[\lambda_{11}(\phi_2), \infty)$, and since $\partial P(\lambda)/\partial\phi < 0$ for $\lambda \geq \lambda_{11}(\phi_2) > 1-\delta$, there cannot be any real roots in $[\lambda_{11}(\phi_2), \infty)$ for $\phi < \phi_2$. Suppose, as ϕ decreases from ϕ_2, at some point a second real root appears in $(1-\delta, \lambda_{11}(\phi_2))$ for the first time at some value $\phi = \tilde{\phi}$. Since this must involve a pair of complex roots colliding and becoming a multiple real root, by Lemma A.7 we must have $P'(\lambda) = 0$ for some $\lambda \in (1-\delta, \lambda_{11}(\phi_2))$ at $\phi = \tilde{\phi}$. But since $\partial P'(\lambda)/\partial\phi < 0$ for $\lambda \in (1-\delta, \lambda_{11}(\phi_2))$ and $P'(\lambda) > 0$ on this interval when $\phi = \phi_2$, we must also have $P'(\lambda) > 0$ for $\phi < \phi_2$, which is a contradiction. Thus, no additional roots ever appear in either $[0, 1-\delta]$, $[\lambda_{11}(\phi_2), \infty)$ or $(1-\delta, \lambda_{11}(\phi_2))$ as ϕ decreases from ϕ_2. Thus, the only remaining possibility is that the next time $\lambda_{11}, \lambda_{12}$ become real they are negative, and by Lemma A.9 this happens at $\phi = \bar{\phi}$. □

A.3.14 Proof of Lemma A.12

By Lemma A.10, all three roots remain real and positive as ϕ decreases from 1 up until λ_{12} collides with λ_2. But this requires one of λ_{12} or λ_2 to have passed through 1, i.e., a fold bifurcation to occur, and since no other type of bifurcation is possible with three real positive roots, the first bifurcation must be a fold. □

A.3.15 Proof of Lemma A.13

The given structure for $\phi = 1$ implies that $P(1-\delta) < 0$, and therefore $\alpha_1 > 0$, $\lambda_2(\phi) > 1-\delta$ for all ϕ, and that $1-\delta$ can never be a root of P for any ϕ. Thus, as long as $\lambda_{11}, \lambda_{12}$ remain real as ϕ decreases from 1, we must have $\lambda_{12} < \lambda_2$ (since otherwise a real root would have to pass through $1-\delta$). If $\lambda_{11}(1) < \lambda_{12}(1)$, then Lemma A.8 implies λ_{11} increases and λ_{12} decreases as ϕ decreases, until they collide, which clearly must happen at a value in $(0, 1-\delta)$. Letting ϕ_0 be the value of ϕ at which this happens, Lemma A.2(iii) implies that there are no roots in $(0, 1-\delta)$ for $\phi < \phi_0$. Since the roots are continuous in ϕ, for ϕ just below ϕ_0, $\lambda_{11}, \lambda_{12}$ become complex. Since we must eventually have $\lambda_{11}, \lambda_{12}$ real and negative by Lemma A.9, these roots must become real again at some value of $\phi < \phi_0$. Let ϕ_1 be the value of ϕ at which this first happens, and $\tilde{\lambda} = \lambda_{12}(\phi_1)$. Since we cannot have a root in $(0, 1-\delta)$ for $\phi < \phi_0$, and since 0 and $1-\delta$ cannot be roots for any ϕ, we either have $\tilde{\lambda} < 0$ (so that $\phi_1 = \bar{\phi}$) or $\tilde{\lambda} > 1-\delta$. □

A.3.16 Proof of Corollary 3

Suppose $\lambda_{12}(1) > 1-\delta$. Then by Lemma A.10, as ϕ decreases from 1, the roots remain real until λ_{12} and λ_2 collide and become complex, and by Lemma A.11 these complex roots persist until $\phi = \bar{\phi}$, and by Lemma A.9 we can never have complex roots again. Thus, there is only one range of ϕ with complex roots for this case.

Suppose instead $\lambda_{12}(1) < 1-\delta$. Then from Lemma A.13, as ϕ decreases from 1, the roots are real until λ_{11} and λ_{12} collide and become complex, and these complex roots persist until either $\phi = \bar{\phi}$, or the roots become real at a value $\tilde{\lambda} > 1-\delta$. If it's the former, then by Lemma A.9 we can never again have complex roots. If it's the latter, then $\lambda_{11}(\phi) > 1-\delta$ at this point, so that, similar to the $\lambda_{11}(1) > 1-\delta$ case, Lemmas A.10 and A.11 imply that the roots become complex exactly one more time as ϕ continues to decrease. Thus, there are at most two intervals where there are complex roots. □

A.3.17 Proof of Lemma A.14

Note that the roots are all continuous in ϕ (subject to the caveat noted in the proof of Lemma A.9). Since $\alpha_1 > 0$, by Lemma A.6 we have $\lambda_2 > 1-\delta$, and by Lemma A.8 this implies that $\lambda_2'(\phi) > 0$ as long as $\lambda_{12} \neq \lambda_2$, which is necessarily true for $\phi \in (\bar{\phi}, \phi_0)$. Since $|\lambda_{12}|^2 = \lambda_{11}\lambda_{12} = (1-\delta)\alpha_2/(\alpha_3\lambda_2)$ on this range of ϕ, this in turn implies that $\partial|\lambda_{12}|/\partial\phi < 0$, and since $\lambda_{12}(\phi)$ is a continuous function and $\phi_1 < \phi_0$, we conclude that $|\lambda_{12}(\phi_1)| > |\lambda_{12}(\phi_0)|$. □

A.3.18 Proof of Lemma A.15

By definition, a hysteresis-inducing fold involves the intermediate of three real roots (i.e., λ_{12} when all roots are real) increasing through 1. A necessary and sufficient condition for such a bifurcation is that (i) there is a ϕ with $P(1) = 0$ (i.e., 1 is a root), and (ii) at that value of ϕ, $P'(1) < 0$ (which, from Lemma A.5, is the condition such that this root is the intermediate of three real ones, i.e., λ_{12}). One can verify that $P(1) = 0$ (i.e., 1 is a root) if and only if $\phi = \phi_0 \equiv \alpha_2 + \alpha_3 - \alpha_1/\delta$, so that condition (i) always holds. Meanwhile, one can verify that $P'(1) < 0$ when $\phi = \phi_0$ if and only if $\alpha_1 < \alpha_1^{**}$. Thus, $\alpha_1 < \alpha_1^{**}$ is necessary and sufficient for a hysteresis-inducing fold to occur, which completes the first part of the proof.

To see that this is the first bifurcation when $\alpha_1 < \alpha_1^{**}$, consider what happens as ϕ decreases from 1 to ϕ_0. Note that, since $\partial P(1)/\partial\phi < 0$, there is only one value of ϕ at which a fold of any kind can occur, which must be $\phi = \phi_0$. Further, since $P(1) = 0$ when $\phi = \phi_0$, for $\phi > \phi_0$ we must have $P(1) < 0$, and therefore $\lambda_2 > 1$.

Now, if $\lambda_{12}(1) > 1 - \delta$, then by Lemma A.12 the first bifurcation is necessarily a fold, and the above implies that this first bifurcation is indeed a hysteresis-inducing fold.

Suppose instead that $\lambda_{12}(1) < 1-\delta$. Note that this implies that $\alpha_1 > 0$ and $P(1-\delta) < 0$. From Lemma A.9, the only possible configurations for $\phi \in (\phi_0, 1)$ are three real positive roots, or one positive and two complex. Thus, if some other bifurcation happens in $(\phi_0, 1)$, it must either be a fold or a Hopf. Since the only fold happens at $\phi = \phi_0$, this leaves only a Hopf as a possibility. Suppose then that a Hopf bifurcation does occur in $(\phi_0, 1)$. Since we have three real roots with $0 < \lambda_{11} < \lambda_{12} \leq 1 < \lambda_2$ for both $\phi = \phi_0$ (since $\alpha_1 < \alpha_1^{**}$) and $\phi = 1$, and since we cannot have a real root pass through 1 as ϕ decreases on $(\phi_0, 1)$, a necessary condition for a Hopf to occur as ϕ decreases from 1 to ϕ_0 is that $\lambda_{11}, \lambda_{12}$ become complex at some $\phi = \tilde{\phi} \in (\phi_0, 1)$ while they are still stable. Let $\phi' \in (\phi_0, \tilde{\phi})$ be the value of ϕ at which they next become real again.

If $\lambda_{12}(\phi') < 0$, then by Lemma A.9 $\lambda_{11}, \lambda_{12}$ remain real and negative for all $\phi < \phi'$. But this contradicts the fact that $0 < \lambda_{11} < \lambda_{12} < 1 < \lambda_2$ for $\phi = \phi_0 < \phi'$, and thus this can't be the case. Suppose instead that $\lambda_{12}(\phi') \in (1-\delta, 1)$, so that a Hopf bifurcation cannot occur in $(\phi', \tilde{\phi})$ (this follows by Lemma A.14, since $|\lambda_{12}|$ strictly increases as ϕ decreases from $\tilde{\phi}$ to ϕ', and $|\lambda_{12}(\phi')| < 1$). Then by Lemma A.10, the next time complex roots appear as ϕ decreases from ϕ' involves λ_{12} and λ_2 colliding. But this implies that one of these roots must have passed through 1, i.e., the first bifurcation is a fold bifurcation, which must be a hysteresis-inducing fold.

Suppose finally that $\lambda_{12}(\phi') > 1$, so that a Hopf bifurcation occurs in $(\phi', \tilde{\phi})$. By Corollary 3, there are at most two disjoint intervals for ϕ where $\lambda_{11}, \lambda_{12}$ are complex. Since one of these intervals is $(\phi', \tilde{\phi})$, and since by Lemma A.9 the other interval must have $\bar{\phi} < \phi'$ as its lower bound, the next time the roots become complex as ϕ decreases from ϕ' is the last. Let $\hat{\phi}$ be the value of ϕ at which this happens. We therefore have $\bar{\phi} < \hat{\phi} < \phi' < \tilde{\phi} < 1$, where $\lambda_{11}, \lambda_{12}$ are negative for $\phi < \bar{\phi}$, complex for $\phi \in (\bar{\phi}, \hat{\phi}) \cup (\phi', \tilde{\phi})$, and positive for $\phi \in (\hat{\phi}, \phi')$. Since $\phi_0 < \phi'$ and $\lambda_{11}(\phi_0), \lambda_{12}(\phi_0)$ are positive, it follows that $\phi_0 \in (\hat{\phi}, \phi')$, and therefore all three roots must be real and positive for $\phi \in [\phi_0, \phi']$. Now, since $\phi' > \phi_0$, we have $P(1) < 0$ for $\phi = \phi'$. Thus, if P has a real root in $(0, 1)$ for $\phi = \phi'$, it must have two. This necessarily requires $\lambda_{12}(\phi') < 1$, which is not the case here. Thus, for $\phi = \phi'$ all three roots must be greater than 1, and in particular $\lambda_{11}(\phi') > 1$. Thus, $\lambda_{11}(\phi)$ is a continuous, real-valued function on $[\phi_0, \phi']$, with $\lambda_{11}(\phi_0) < 1 < \lambda_{11}(\phi')$, which implies that we must have $\lambda_{11}(\phi_1) = 1$ for some $\phi_1 \in (\phi_0, \phi')$. But $P(0) < 0$ for $\phi > \phi_0$, and therefore this is not possible, which is a contradiction. We therefore conclude that we cannot have $\lambda_{12}(\phi') > 1$ in this case, and therefore a Hopf bifurcation will not occur. This completes the proof. □